SARANATHAN COLLEGE OF ENGINEERING, TIRUCHIRAPPALLI - 620012

(AN AUTONOMOUS INSTITUTION)

(Affiliated to Anna University, Chennai)
Venkateswara Nagar, Panjappur
Tiruchirappalli – 620012

B.E./B.TECH. REGULATIONS 2024

B.E. COMPUTER SCIENCE AND ENGINEERING

CURRICULUM AND SYLLABUS
(For the students Admitted from the Academic Year 2024 onwards)

Version: 1.0	**Date: 28-08-2024**

SARANATHAN COLLEGE OF ENGINEERING, TIRUCHIRAPPALLI - 620012
AN AUTONOMOUS INSTITUTION
REGULATIONS 2024

B.E. COMPUTER SCIENCE AND ENGINEERING

CHOICE BASED CREDIT SYSTEM

PROGRAM EDUCATIIONAL OBJECTIVES (PEOs)

Graduates of Computer Science and Engineering will be able to

- Acquire strong foundation in the mathematical, scientific and engineering fundamentals necessary to formulate, solve and analyze engineering problems.
- Develop the ability to analyze the requirements of the software, understand the technical specifications, design and provide novel engineering solutions and efficient software/hardware designs.
- Have exposure to emerging cutting edge technologies, adequate training & opportunities to work as teams on multidisciplinary projects with effective communication skills and leadership qualities.
- Have awareness on the life-long learning and prepare them for research development and consultancy.
- Have a successful career and work with values & social concern bridging the digital divide and meet the requirements of Indian and multinational companies.

PROGRAM OUTCOMES (POs)

Engineering Graduates will be able to

1. **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.
2. **Problem analysis:** Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.
3. **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety and the cultural, societal and environmental considerations.
4. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data and synthesis of the information to provide valid conclusions.
5. **Modern tool usage:** Create, select, and apply appropriate techniques, resources and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
6. **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

7. **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts and demonstrate the knowledge of and need for sustainable development.
8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9. **Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
10. **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large such as being able to comprehend and write effective reports and design documentation, make effective presentations and give and receive clear instructions.
11. **Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team to manage projects and in multidisciplinary environments.
12. **Life-long learning:** Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs)

The Students will be able to

1. **Foundation of mathematical concepts:** To use mathematical methodologies to crack problem using suitable mathematical analysis, data structure and suitable algorithm.
2. **Foundation of Computer System:** the ability to interpret the fundamental concepts and methodology of computer systems. Students can understand the functionality of hardware and software aspects of computer systems.
3. **Foundations of Software development:** the ability to grasp the software development lifecycle and methodologies of software systems. Possess competent skills and knowledge of software design process. Familiarity and practical proficiency with a broad area of programming.

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CHOICE BASED CREDIT SYSTEM
B.E. COMPUTER SCIENCE AND ENGINEERING
CURRICULUM AND SYLLABUS FOR SEMESTERS I to IV

SEMESTER I								
S. NO.	COURSE CODE	COURSE TITLE	CATEGORY	TOTAL CONTACT PERIODS	PERIOD PER WEEK			CREDITS
					L	T	P	
1	24IP101	Induction Programme	-	-	-			0
THEORY (IA: 40 & ESE: 60)								
2	24EN101	Professional English I	HSMC	3	3	0	0	3
3	24MA101	Engineering Mathematics I	BSC	4	3	1	0	4
4	24PH101	Engineering Physics	BSC	3	3	0	0	3
5	24CH101	Engineering Chemistry	BSC	3	3	0	0	3
6	24ES101	Problem Solving and Python Programming	ESC	3	3	0	0	3
7	24ES102	Engineering Graphics	ESC	6	2	0	4	4
8	24TA101	தமிழர் மரபு /Heritage of Tamils	HSMC	1	1	0	0	1
PRACTICALS (IA: 60 & ESE: 40)								
9	24PH111	Physics and Chemistry Laboratory	BSC	4	0	0	4	2
10	24ES111	Python Programming Laboratory	ESC	4	0	0	4	2
11	24EM111	Communication Laboratory I	EEC	2	0	0	2	1
			TOTAL	**33**	**18**	**1**	**14**	**26**

SEMESTER II								
S. NO.	COURSE CODE	COURSE TITLE	CATEGORY	TOTAL CONTACT PERIODS	PERIOD PER WEEK			CREDITS
					L	T	P	
THEORY (IA: 40 & ESE: 60)								
1	24EN201	Professional English II	HSMC	3	3	0	0	3
2	24MA201	Engineering Mathematics II	BSC	4	3	1	0	4
3	24PH201	Physics for Information Science	BSC	3	3	0	0	3
4	24CS201	Programming in C	PCC	3	3	0	0	3
5	24TA201	தமிழரும் தொழில்நுட்பமும்/ Tamils and Technology	HSMC	1	1	0	0	1
THEORY CUM PRACTICAL (IA: 50 & ESE: 50)								
6	24ES201	Basic Electrical and Electronics Engineering	ESC	5	3	0	2	4
PRACTICALS (IA: 60 & ESE: 40)								
7	24CS211	Programming in C Laboratory	PCC	4	0	0	4	2
8	24EM211	Communication Laboratory II	EEC	2	0	0	2	1
			TOTAL	**25**	**16**	**1**	**8**	**21**

SEMESTER III								
S.NO.	**COURSE CODE**	**COURSE TITLE**	**CATEGORY**	**TOTAL CONTACT PERIODS**	**PERIOD PER WEEK**			**CREDITS**
					L	**T**	**P**	
THEORY (IA: 40 & ESE: 60)								
1	24MA301	Probability and Statistics	BSC	4	3	1	0	4
2	24CS301	Data Structures	PCC	3	3	0	0	3
3	24CS302	Object Oriented Programming	PCC	3	3	0	0	3
THEORY CUM PRACTICAL (IA: 50 & ESE: 50)								
4	24ES301	Digital Principles and Computer Organization	ESC	5	3	0	2	4
5	24CS303	Essentials of Web Development	PCC	5	3	0	2	4
6	24CS304	Foundations of Data Science	PCC	5	3	0	2	4
PRACTICALS (IA: 60 & ESE: 40)								
7	24CS311	Data Structures Laboratory	PCC	3	0	0	3	1.5
8	24CS312	Object Oriented Programming Laboratory	PCC	3	0	0	3	1.5
9	24EM301	Employability Skills I	EEC	2	0	0	2	1
			TOTAL	**33**	**18**	**1**	**14**	**26**

SEMESTER IV								
S.NO.	**COURSE CODE**	**COURSE TITLE**	**CATEGORY**	**TOTAL CONTACT PERIODS**	**PERIOD PER WEEK**			**CREDITS**
					L	**T**	**P**	
THEORY (IA: 40 & ESE: 60)								
1	24MA401	Discrete Mathematics	BSC	4	3	1	0	4
2	24CH401	Environmental Sciences and Sustainability	BSC	2	2	0	0	2
3	24CS401	Theory of Computation	PCC	3	3	0	0	3
4	24CS402	Database Management Systems	PCC	3	3	0	0	3
5	24CS403	Operating Systems	PCC	3	3	0	0	3
THEORY CUM PRACTICAL (IA: 50 & ESE: 50)								
6	24CS404	Design and Analysis of Algorithms	PCC	5	3	0	2	4
PRACTICALS (IA: 60 & ESE: 40)								
7	24CS411	Database Management Systems Laboratory	PCC	3	0	0	3	1.5
8	24CS412	Operating Systems Laboratory	PCC	3	0	0	3	1.5
9	24EM401	Employability Skills II	EEC	2	0	0	2	1
10		NSS / YRC Non-Credit Course		-	-	-	-	-
			TOTAL	**28**	**17**	**1**	**10**	**23**

SCE-R 2024/V2.0|Approved in 2nd Academic Council Meeting held on 30.06.25

MAPPING OF COURSE OUTCOME AND PROGRAM OUTCOME (POs)																	
YEAR	**SEM**	**COURSE TITLE**	**PROGRAM OUTCOMES (POs)**												**PSOs**		
			1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**1**	**2**	**3**
I	I	Professional English I	-	-	-	-	-	-	-	2	3	3	1	1	-	-	-
		Engineering Mathematics I	3	3	2	2	-	-	-	-	2	-	-	2	-	-	-
		Engineering Physics	3	2	2	1	2	1	-	-	-	-	-	1	-	-	-
		Engineering Chemistry	3	2	2	1	-	2	2	-	-	-	-	1	-	-	-
		Problem Solving and Python Programming	3	3	2	2	2	-	-	-	1	-	1	1	3	2	1
		தமிழர் மரபு /Heritage of Tamils	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
		Engineering Graphics	3	1	2	-	2	-	-	-	-	3	-	2	2	2	-
		Physics and Chemistry Laboratory	3	2	2	1	2	2	-	-	-	-	-	-	-	-	3
		Python Programming Laboratory	3	3	2	2	2	-	-	-	2	-	1	1	3	2	1
		Communication Laboratory I	-	-	-	1	-	1	1	2	3	3	1	2	-	-	-
I	II	Professional English II	-	-	-	-	-	-	-	2	3	3	1	1	-	-	-
		Engineering Mathematics II	3	3	2	2	-	-	-	-	2	-	-	2	-	-	-
		Physics for Information Science	3	1	2	2	2	1	2	-	-	-	-	2	-	-	-
		Programming in C	3	3	2	2	1	-	-	-	1	-	1	1	3	2	1
		தமிழரும் தொழில்நுட்பமும்/ Tamils and Technology	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
		Basic Electrical and Electronics Engineering	3	3	3	2	2	1	-	-	2	-	2	2	2	2	3
		Programming in C Laboratory	3	3	2	2	1	-	-	-	2	-	1	1	3	2	1
		Communication Laboratory II	-	-	-	-	-	-	-	2	3	3	-	1	-	-	-

1 - low, 2 - medium, 3 - high, '-' - no correlation

24IP101 INDUCTION PROGRAMME

This is a mandatory 2 week programme to be conducted as soon as the students enter the institution. Normal classes start only after the induction program is over.

The induction programme has been introduced by AICTE with the following objective:

“Engineering colleges were established to train graduates well in the branch/department of admission, have a holistic outlook, and have a desire to work for national needs and beyond. The graduating student must have knowledge and skills in the area of his/her study. However, he/she must also have a broad understanding of society and relationships. Character needs to be nurtured as an essential quality by which he/she would understand and fulfil his/her responsibility as an engineer, a citizen and a human being. Besides the above, several meta-skills and underlying values are needed.”

“One will have to work closely with the newly joined students in making them feel comfortable, allow them to explore their academic interests and activities, reduce competition and make them work for excellence, promote bonding within them, build relations between teachers and students, give a broader view of life, and build character.”

Hence, the purpose of this programme is to make the students feel comfortable in their new environment, open them up, set a healthy daily routine, create bonding in the batch as well as between faculty and students, develop awareness, sensitivity and understanding of the self, people around them, society at large, and nature.

The following are the activities under the induction program in which the student would be fully engaged throughout the day for the entire duration of the program.

(i) Physical Activity

This would involve a daily routine of physical activity with games and sports, yoga, gardening, etc.

(ii) Creative Arts

Every student would choose one skill related to the arts whether visual arts or performing arts. Examples are painting, sculpture, pottery, music, dance etc. The student would pursue it everyday for the duration of the program. These would allow for creative expression. It would develop a sense of aesthetics and also enhance creativity which would, hopefully, grow into engineering design later.

(iii) Universal Human Values

This is the anchoring activity of the Induction Programme. It gets the student to explore oneself and allows one to experience the joy of learning, stand up to peer pressure, make decisions with courage, be aware of relationships with colleagues and supporting stay in the hostel and department, be sensitive to others, etc. A module in Universal Human Values provides the base. Methodology of teaching this content is extremely important. It must not be through do's and don’ts but get students to explore and think by engaging them in a dialogue. It is best taught through group discussions and real life activities rather than lecturing.
Discussions would be conducted in small groups of about 20 students with a faculty 23 mentor each. It would be effective that the faculty mentor assigned is also the faculty advisor for the student for the full duration of the UG programme.

(iv) Literary Activity

Literary activity would encompass reading, writing and possibly, debating, enacting a play etc.

(v) Proficiency Modules

This would address some lacunas that students might have, for example, English, computer familiarity etc.

(vi) Lectures by Eminent People

Motivational lectures by eminent people from all walks of life should be arranged to give the students exposure to people who are socially active or in public life.

(vii) Visits to Local Area

A couple of visits to the landmarks of the city, or a hospital or orphanage could be organized. This would familiarize them with the area as well as expose them to the underprivileged.

(viii) Familiarization to Dept./Branch & Innovations

They should be told about what getting into a branch or department means what role it plays in society, through its technology. They should also be shown the laboratories, workshops & other facilities.

(ix) Department Specific Activities

About a week can be spent in introducing activities (games, quizzes, social interactions, small experiments, design thinking etc.) that are relevant to the particular branch of Engineering / Technology / Architecture that can serve as a motivation and kindle interest in building things (become a maker) in that particular field. This can be conducted in the form of a workshop. For example, CSE and IT students may be introduced to activities that kindle computational thinking, and get them to build simple games. ECE students may be introduced to building simple circuits as an extension of their knowledge in Science, and so on. Students may be asked to build stuff using their knowledge of science.

Induction Programme is totally an activity based programme and therefore there shall be no tests / assessments during this programme.

REFERENCES:

Guide to Induction program from AICTE

SEMESTER-WISE CREDITS DISTRIBUTION

SUMMARY											
Sl. No.	**Course Category**	**Credits per Semester**								**Credits**	**Credit %**
		I	**II**	**III**	**IV**	**V**	**VI**	**VII**	**VIII**		
1	HSMC	4	4	-	-						
2	BSC	12	7	4	6						
3	ESC	9	4	4	-						
4	EEC	1	1	1	1						
5	PC	-	5	17	16						
6	PE	-	-	-	-						
7	OE	-	-	-	-						
8	PW	-	-	-	-						
9	MC/NC/AC	-	-	-	-						
	Total	**26**	**21**	**26**	**23**						

TYPE	TYPE OF CONTACT	CAT	CATEGORY OF COURSE	CAT	CATEGORY OF COURSE
L	Lecture Period	HSMC	Humanities, Social Sciences and Management Courses	PW	Project Work Courses
T	Tutorial Period	BSC	Basic Science Courses	EEC	Employability Enhancement Courses
P	Laboratory Period	ESC	Engineering Science Courses	MC/NC/AC	Mandatory Courses/Non-Credit Courses/Audit Courses
C	Credits	PC	Professional Core Courses	**MODE OF ASSESSMENT**	
		PE	Professional Elective Courses	IA	Internal Assessment
		OE	Open Elective Courses	ESE	End Semester Examination

24EN101 **PROFESSONAL ENGLISH I**
Offered by English (Common to ALL branches) (I Sem.)

L	T	P	C
3	0	0	3

COURSE OBJECTIVES

- To improve the communicative competence of learners
- To learn and apply basic grammatical structures in suitable contexts
- To acquire lexical competence and use them appropriately in a sentence and understand their meaning in a text like recommendation, instruction
- To help learners use language effectively in transferring data / graphs
- To develop learners' ability to read and write complex texts, summaries, articles, blogs, definitions, essays and user manuals
- To develop and demonstrate basic communication skills in technical and professional contexts effectively

UNIT I FUNDAMENTALS OF GRAMMMAR **9**

Reading - Reading short texts from journals - Newspapers, reading novels. **Writing** - Writing formal letters and emails. **Grammar** - Parts of Speech - Tenses - Question types: WH-/ Yes or No questions and Tags. **Vocabulary** - Word formation - Prefixes - Suffixes - Changing word from one form to another form - Word used as noun and verb - Singular - Plural.

UNIT II GRAMMAR AND USAGE **9**

Reading - Reading biographies, travelogues, newspaper reports, excerpts from literature, travel and technical blogs. **Writing** - Paragraph writing - Free writing on any given topics (my favourite places, hobbies, school life etc.,) - Process / Product description. **Grammar** - Subject-Verb agreement - Gerund and Infinitive - Discourse markers (connectives & sequence words) - Reference words. **Vocabulary** - Synonyms - Antonyms - One word substitution - Abbreviations & Acronyms (as used in technical contexts).

UNIT III FORMAL WRITING **9**

Reading - Reading advertisements, gadget reviews, user manuals. **Writing** - Writing definitions - Instructions - Recommendations. **Grammar** - Imperatives - Adjectives - Degrees of comparison. **Vocabulary** - Misspelt words - Commonly confused words.

UNIT IV TRANSFERRING INFORMATION **9**

Reading - Interpreting visual materials (Bar Chart, Pie Chart, Table, Flow chart). **Writing** - Note- making / Note-taking transferring information from non-verbal (chart, graph etc., to verbal mode) **Grammar** - Articles, Pronouns - Possessive & Relative pronouns, relative clause, prepositions, error correction. **Vocabulary** - Collocations - Compound words - Fixed / Semi- fixed expressions.

UNIT V WRITING ESSAY **9**

Reading - Reading comprehension. **Writing** - Essay Writing (descriptive, narrative, argumentative, cause and effect essays). **Grammar** - Spelling and punctuation, Sentence formation - Negation (Statements & Questions) - Simple, Compound & Complex sentences. **Vocabulary** - Cause & Effect expressions, verbal analogies, expose to different Jargons.

TOTAL: 45 PERIODS

TEXT BOOKS

1. Department of English, "English for Engineers & Technologists", Anna University, Orient Black swan Private Ltd, 2020.
2. Dr. Veena Selvam, Dr. Sujatha Priyadarshini, Dr. Deepa Mary Francis, Dr. KN.Shoba, and Dr. Lourdes Joevani, "English for Science & Technology", Cambridge University Press, 2021.

REFERENCES

1. Meenakshi Raman & Sangeeta Sharma, "Technical Communication – Principles and Practices", Oxford Univ. Press, New Delhi. 2016.
2. Lakshminarayanan, "A Course Book on Technical English", Scitech Publications, India.

3. Aysha Viswamohan, "English for Technical Communication" (With CD), Mcgraw Hill Education, ISBN : 0070264244.
4. Kulbhusan Kumar, RS Salaria, "Effective Communication Skill", Khanna Publishing House.
5. Dr. V. Chellammal, "Learning to Communicate," Allied Publishing House, New Delhi, 2003.

COURSE OUTCOMES

After the completion of the course, the students will be able to

CO1: To use appropriate words in a professional context

CO2: To gain understanding of basic grammatical structures and use them in right context

CO3: To read and infer the denotative and connotative meanings of technical texts

CO4: To write definitions, descriptions, narrations and essays on various topics

CO5: To improve reading skill and comprehend

CO6: To showcase the writing skill through various types of essays

CO's-PO's & PSO's MAPPING

COs	POs												PSOs		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
1	-	-	-	-	-	-	-	2	3	3	1	1	-	-	-
2	-	-	-	-	-	-	-	2	3	3	1	1	-	-	-
3	-	-	-	-	-	-	-	2	3	3	1	1	-	-	-
4	-	-	-	-	-	-	-	2	3	3	1	1	-	-	-
5	-	-	-	-	-	-	-	2	3	3	1	1	-	-	-
6	-	-	-	-	-	-	-	2	3	3	1	1	-	-	-
Avg.	-	-	-	-	-	-	-	2	3	3	1	1	-	-	-

1 - Low, 2 - medium, 3 - high, '-"- no correlation

24MA101 ENGINEERING MATHEMATICS I

Offered by Maths (Common to ALL branches) (I Sem.)

L	T	P	C
3	1	0	4

COURSE OBJECTIVES

- To develop the ability to use matrix algebra techniques for practical applications.
- To provide a clear understanding of the limit and continuity of a function.
- To educate students on differential calculus.
- To familiarize students with functions of several variables, which are essential in various engineering fields.
- To help students understand different techniques of integration.
- To acquaint students with mathematical tools necessary for evaluating multiple integrals and their applications.

UNIT I MATRICES 9+3

Eigen values and Eigenvectors of a real matrix – Characteristic equation – Properties of Eigenvalues and Eigenvectors – Cayley-Hamilton theorem – Diagonalization of matrices by orthogonal transformation – Reduction of a quadratic form to canonical form by orthogonal transformation – Nature of quadratic forms.

UNIT II DIFFERENTIAL CALCULUS 9+3

Limit of a function – Continuity – Derivatives – Differentiation rules – sum, product, quotient and chain rules – Implicit differentiation–Logarithmic differentiation – Applications: Maxima and Minima of functions of one variable.

UNIT III FUNCTIONS OF SEVERAL VARIABLES 9+3

Partial differentiation – Homogeneous functions and Euler's theorem – Total derivative – Change of variables – Jacobians – Partial differentiation of implicit functions – Taylor's series for functions of two variables – Applications: Maxima and minima of functions of two variables and Lagrange's method of undetermined multipliers.

UNIT IV SINGLE INTEGRAL 9+3

Definite and Indefinite integrals – Substitution rule – Techniques of Integration: Integration by parts, Trigonometric integrals, Integration of rational functions by partial fraction, Integration of irrational functions – Improper integrals.

UNIT V MULTIPLE INTEGRALS 9+3

Double integrals – Change of order of integration – Double integrals in polar coordinates – Area enclosed by plane curves – Change of variables in double integrals – Triple integrals – Volume of solids

TOTAL: 60 PERIODS

TEXT BOOKS

1. ErwinKreyszig,"AdvancedEngineeringMathematics",JohnWiley&Sons,10thEdition,2017.
2. Grewal, B.S.,"Higher Engineering Mathematics ",Khanna Publishers, NewDelhi, 44th Edition,2021.
3. James Stewart, Daniel Clegg, K., Saleem Watson, Lothar Redlin, "Calculus: Early Transcendentals", Cengage Learning Publishers, NewDelhi, 9th Edition,2020.

REFERENCES

1. Howard Anton, Irl Bivens, Stephen Davis, "Calculus", John Wiley & Sons,12thEdition, 2021.
2. Bali,N.P.,Goyal,M.,Watkins,C.,"Advanced Engineering Mathematics: A Complete Approach", Laxmi Publications,7thEdition, 2015.
3. Jain,R.K., and Iyengar,S.R.K., "Advanced Engineering Mathematics", Narosa Publications, NewDelhi, 9threprint, 2020.

4. Narayanan,S. and Manicavachagom Pillai,T.K.," Calculus- Volume I and II", S. Viswanathan Publishers Pvt. Ltd.,Chennai, 2009.
5. Srimantha Paland Bhunia, S.C., "Engineering Mathematics", Oxford University Press, 2015.
6. MauriceD.Weir, JoelHass, ChristopherHeil, PrzemyslawBogacki, "Thomas' Calculus", Pearson Publications, 15th Edition, 2024.

COURSE OUTCOMES

After the completion of the course, the students will be able to

CO1: Apply matrix algebra methods to solve practical problems.

CO2: Comprehend the concepts of limits and continuity of functions.

CO3: Utilize differentiation rules to solve problems in various applications.

CO4: Analyze differential calculus to functions of several variables.

CO5: Acquire knowledge of different integration methods to solve practical problems.

CO6: Implement multiple integrals to solve problems involving areas, volumes, and other practical applications.

CO's-PO's & PSO's MAPPING

COs	POs												PSOs		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
1	3	3	2	2	-	-	-	-	2	-	-	2	-	-	-
2	3	3	2	2	-	-	-	-	2	-	-	2	-	-	-
3	3	3	2	2	-	-	-	-	2	-	-	2	-	-	-
4	3	3	2	2	-	-	-	-	2	-	-	2	-	-	-
5	3	3	2	2	-	-	-	-	2	-	-	2	-	-	-
6	3	3	2	2	-	-	-	-	2	-	-	2	-	-	-
Avg.	3	3	2	2	-	-	-	-	2	-	-	2	-	-	-

1 - Low, 2 - medium, 3 - high, '-"- no correlation

24PH101 ENGINEERING PHYSICS

Offered by Physics (Common to ALL branches) (I Sem.)

L	T	P	C
3	0	0	3

COURSE OBJECTIVES

- To acquire the fundamental concepts in mechanics and moment of inertia.
- To understand the fundamental concepts of oscillations and waves.
- To know the properties of electromagnetic waves.
- To gain knowledge in quantum mechanics and its applications.
- To learn the fundamental concepts of laser and its applications.
- To attain the basic knowledge of fiber optic techniques.

UNIT I MECHANICS **9**

Multi-particle dynamics – Center of mass (CM) – CM of continuous bodies – Motion of the CM –Kinetic energy of the system of particles – Rotation of rigid bodies: rotational kinetic energy and moment of inertia (M.I) – Theorems of M.I – Moment of inertia of continuous bodies (thin rod, circular disc and solid sphere) – Moment of inertia of diatomic molecule – Rotational dynamics of rigid bodies – Gyroscope – Torsional pendulum.

UNIT II OSCILLATIONS AND WAVES **9**

Simple harmonic motion – Resonance – Analogy between mechanical and electrical oscillating systems – Standing waves – Travelling waves – Sound waves – Doppler effect – applications of Doppler effect – Interference of light–Air wedge (theory& experiment).

UNIT III ELECTRO MAGNETIC WAVES **9**

Introduction – Electric flux density – Electric field Intensity – Magnetic flux density – Magnetic field Intensity – Electric and Magnetic Gauss' Law – Faraday's Law – Electric Displacement vector – Maxwell's equations – Plane electromagnetic waves in vacuum – properties of electromagnetic waves: velocity, amplitude, phase and orientation – Polarization – Production of electromagnetic waves – Momentum and radiation pressure – Cell phone reception.

UNIT IV QUANTUM MECHANICS **9**

Introduction – Dual nature of electron – de-Broglie waves – Compton effect (theory & experimental verification) – Schrodinger's equation (time dependent and independent) – meaning of wave function – Normalization – free particle – particle in a infinite potential well: 1D box – Barrier penetration and quantum tunneling – Scanning tunneling microscope – Bloch theorem – Kronig-Penney model.

UNIT V LASER AND FIBER OPTICS **9**

Laser: Characteristics – Stimulated absorption, spontaneous emission and stimulated emission – Einstein's A and B coefficients–derivation – population inversion – pumping methods – resonant cavity–Nd-YAG Laser–Carbon dioxide laser – Applications of laser.
Fiber optics – Principle, numerical aperture and acceptance angle – Types of optical fiber: material, mode and refractive index –attenuation associated with optical fiber.

TOTAL: 45 PERIODS

TEXT BOOKS

1. B.K.Pandey and S.Chaturvedi, "Engineering Physics", Cengage Learning India,2017.
2. M.N. Avadhanulu and P.G. Kshirshagar, "A Textbook of Engineering Physics", S. Chand & Co Ltd. 2016.
3. D.K.Bhattacharya and T.Poonam, "Engineering Physics", Oxford University Press, 2015.

REFERENCES

1. R.Wolfson, "Essential University Physics", Volume no.1 & 2, Pearson Education (Indian Edition), 2020.
2. K.Thyagarajan and A.Ghatak, "Lasers - Fundamentals and Applications", Laxmi Publications, (Indian Edition), 2019.
3. D.Halliday, R.Resnick and J.Walker," Principles of Physics", Wiley (Indian Edition), 2015.

4. N.Garcia, A.Damask and S.Schwarz, “Physics for Computer Science Students: With emphasis on Atomic and Semiconductor Physics”, Springer-Verlag, 2012.

COURSE OUTCOMES

After the completion of the course, the students will be able to

CO1: Know the importance of mechanics used in engineering and technology.

CO2: Acquire the fundamental knowledge of oscillations and waves.

CO3: Obtain the knowledge of electromagnetic waves and their properties.

CO4: Understand the knowledge in basic quantum mechanics and its applications.

CO5: Learn the fundamental concepts of laser and its applications.

CO6: Gain the basic knowledge of fiber optics.

CO’s-PO’s & PSO’s MAPPING

COs	POs												PSOs		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
1	3	2	2	1	2	1	-	-	-	-	-	1	-	-	-
2	3	2	2	1	2	1	-	-	-	-	-	1	-	-	-
3	3	2	2	1	2	1	-	-	-	-	-	1	-	-	-
4	3	2	2	1	2	1	-	-	-	-	-	1	-	-	-
5	3	2	2	1	2	1	-	-	-	-	-	1	-	-	-
6	3	2	2	1	2	1	-	-	-	-	-	1	-	-	-
Avg.	3	2	2	1	2	1	-	-	-	-	-	1	-	-	-

1 - Low, 2 - medium, 3 - high, ‘-“- no correlation

24CH101 ENGINEERING CHEMISTRY

Offered by Chemistry (Common to ALL branches) (I Sem.)

L	T	P	C
3	0	0	3

COURSE OBJECTIVES

- To make the students conversant with quality of boiler feed water, its related problems and water treatment techniques.
- To illustrate the principles of electro chemical reactions, redox reactions in corrosion of materials and methods for corrosion prevention.
- To demonstrate the principles and generation of energy in batteries, nuclear reactors and fuel cells.
- To assimilate the preparation, properties and applications of nanomaterials in various fields.
- To categorize types of fuels, manufacture of solid, liquid and gaseous fuels.
- To understand the calorific value calculations.

UNIT I WATER AND ITS TREATMENT 9

Hardness of water: types, expression of hardness, units – Boiler troubles: scale and sludge, priming and foaming, boiler corrosion, caustic embrittlement – Treatment of boiler feed water: Internal treatment – phosphate, colloidal, aluminate and Calgon conditioning – External treatment –ion exchange process, zeolite process – Desalination of brackish water: reverse osmosis.

UNIT II ELECTRO CHEMISTRY AND CORROSION 9

Electrochemical cell – redox reaction, electrode potential – origin of electrode potential – oxidation potential – reduction potential, measurement and applications – electrochemical series and its significance – Nernst equation (derivation and problems).

Corrosion: causes – types – chemical corrosion, Pilling-Bedworth rule – electrochemical corrosion: mechanism – galvanic, differential aeration – Corrosion control: material selection and design aspects, cathodic protection method, sacrificial anode method.

UNIT III ENERGY SOURCES AND STORAGE DEVICES 9

Energy sources: Nuclear fission, nuclear fusion, characteristics, nuclear energy, chain reactions, types of reactors: light water nuclear power plant, breeder reactor, thorium fast reactor.

Storage devices: Batteries, types of batteries, primary battery (dry cell), secondary battery–NICAD, lead acid battery, lithium–ion battery, Fuel cells – H2O2 fuel cell, microbial fuel cell, Super capacitors.

UNIT IV NANOMATERIALS 9

Nanomaterials – nanoparticles, nanoclusters, nanorods, nanotubes (CNT: SWNT and MWNT) and nanowires – Properties – physical – surface to volume ratio, melting point, optical and electrical properties, Synthesis–electros pinning, electrode position, chemical vapour deposition, laser ablation, Applications –medicine, energy, electronics and catalysis.

UNIT V FUELS AND COMBUSTION 9

Fuels: Introduction, classification of fuels – Solid fuels: coal, analysis of coal - proximate and ultimate, carbonization, manufacture of metallurgical coke–Otto Hoffmann method – Liquid fuels: petroleum, manufacture of synthetic petrol (Bergius process), knocking, octane number and cetane number, Gaseous fuels – compressed natural gas (CNG), liquefied petroleum gases (LPG), Biofuels–power alcohol and biodiesel. Combustion of Fuels: Introduction, calorific value, higher and lower calorific values, problems – theoretical calculation of calorific value.

TOTAL: 45 PERIODS

TEXT BOOKS

1. S.S.Dara and S.S.Umare, "A Text book of Engineering Chemistry", S.Chand & Company Ltd, New Delhi, 2015.
2. P.C.Jain and Monika Jain, "Engineering Chemistry", Dhanpat Rai Publishing Company (P) Ltd, New Delhi, 2015.
3. S.Vairam,P.Kalyani and Suba Ramesh, "Engineering Chemistry", Wiley India PVT Ltd, New Delhi, 2013.
4. A.Ravikrishnan, "Engineering Chemistry", Sri Krishna Hitech Publishing Company PVT Ltd, New Edition 2021.

REFERENCES

1. O.G.Palanna, "Engineering Chemistry", Mc Graw Hill Education (India) Private Limited, 2nd Edition, 2017.
2. Friedrich Emich, "Engineering Chemistry", Scientific International PVT LTD, New Delhi,2014.
3. Shikha Agarwal, "Engineering Chemistry - Fundamentals and Applications", Cambridge University Press, New Delhi, 2nd Edition, 2019.
4. M.A.Shah, Tokeer Ahmad, "Principles of nano science and nano technology", Narosa Publishing House PVT Ltd.,2013.
5. O.V.Roussak and H.D.Gesser, "Applied Chemistry-A Text Book for Engineers and Technologists", Springer Science Business Media, NewYork, 2ndEdition,2013.

COURSE OUTCOMES

After the completion of the course, the students will be able to

CO1: To infer the quality of water from quality parameter data and propose suitable treatment methodologies to treat water.

CO2: To gain insights into the basic concepts of electro chemistry and implement its applications in corrosion prevention.

CO3: To recognize different forms of energy resources and apply them for suitable applications in energy sectors.

CO4: To identify and apply basic concepts of nanoscience and nanotechnology in designing the synthesis of nanomaterials for engineering and technology applications.

CO5: To recommend suitable fuels for engineering processes and applications.

CO6: To analyze combustion process and its calculations.

CO's-PO's & PSO's MAPPING

COs	POs												PSOs		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
1	3	2	2	1	-	2	2	-	-	-	-	1	-	-	-
2	3	2	2	1	-	2	2	-	-	-	-	1	-	-	-
3	3	2	2	1	-	2	2	-	-	-	-	1	-	-	-
4	3	2	2	1	-	2	2	-	-	-	-	1	-	-	-
5	3	2	2	1	-	2	2	-	-	-	-	1	-	-	-
6	3	2	2	1	-	2	2	-	-	-	-	1	-	-	-
Avg.	3	2	2	1	-	2	2	-	-	-	-	1	-	-	-

1 - Low, 2 - medium, 3 - high, '-"- no correlation

24ES101 PROBLEM SOLVING AND PYTHON PROGRAMMING

Offered by CSE (Common to ALL branches) (I Sem.)

L	T	P	C
3	0	0	3

COURSE OBJECTIVES

- To understand the basics of algorithmic problem solving.
- To learn to solve problems using Python conditionals and loops.
- To define Python functions and use function calls to solve problems.
- To use Python data structures - lists, tuples, dictionaries to represent complex data.
- To do input/output with files in Python.

UNIT I COMPUTATIONAL THINKING AND PROBLEM SOLVING 9

Fundamentals of computing – identification of computational problems - algorithms, building blocks of algorithms (statements, state, control flow, functions), notation (pseudo code, flow chart, programming language), algorithmic problem solving. Simple strategies for developing algorithms (iteration, recursion). Illustrative problems: find minimum in a list, insert a card in a list of sorted cards, guess an integer number in a range, Towers of Hanoi.

UNIT II DATA TYPES, EXPRESSIONS, STATEMENTS 9

Python interpreter and interactive mode, debugging; values and types: int, float, boolean, string, and list; variables, expressions, statements, tuple assignment, precedence of operators, comments; Illustrative programs: exchange the values of two variables, circulate the values of n variables, distance between two points.

UNIT III CONTROL FLOW, FUNCTIONS, STRINGS 9

Conditionals: Boolean values and operators, conditional (if), alternative (if-else), chained conditional (if-elif-else); Iteration: state, while, for, break, continue, pass; Fruitful functions: return values, parameters, local and global scope, function composition, recursion; Strings: string slices, immutability, string functions and methods, string module; Lists as arrays.

UNIT IV LIST, TUPLE, DICTIONARY 9

Lists: list operations, list slices, list methods, list loop, mutability, aliasing, cloning lists, list parameters; Tuple: tuple assignment, tuple as return value; Dictionary: operations and methods; advanced list processing - list comprehension.

UNIT V FILES, MODULES, PACKAGES 9

Files and exceptions: text files, reading and writing files, format operator; command line arguments, errors and exceptions, handling exceptions, modules, packages.

TOTAL: 45 PERIODS

TEXT BOOKS

1. S.Sridhar, J. Indumathi, V.M. Hariharan, “Python Programming”, Pearson Education, 2023.

REFERENCES

1. Paul Deitel and Harvey Deitel, “Python for Programmers”, Pearson Education, 1st Edition, 2021.
2. G Venkatesh and Madhavan Mukund, “Computational Thinking: A Primer for Programmers and Data Scientists”, 1st Edition, Notion Press, 2021.
3. John V Guttag, "Introduction to Computation and Programming Using Python: With Applications to Computational Modeling and Understanding Data‘‘, 3rd Edition, MIT Press, 2021.
4. R. Nageswara Rao, “Core Python Programming”, 3rd Edition, Dreamtech Press, 2021.

5. Allen B. Downey, “Think Python: How to Think Like a Computer Scientist”, 2nd Edition, O'Reilly, 2016.
6. Ashok Namdev Kamthane and Amit Ashok Kamthane, “Python Programming”, McGraw Hill Education (India) Private Limited, 2018.

COURSE OUTCOMES

After the completion of the course, the students will be able to

CO1: Develop algorithmic solutions to simple computational problems.

CO2: Develop and execute simple Python programs.

CO3: Write simple Python programs using conditionals and loops for solving problems.

CO4: Decompose a Python program into functions.

CO5: Represent compound data using Python lists, tuples, dictionaries etc.

CO6: Read and write data from/to files in Python programs.

CO’s-PO’s & PSO’s MAPPING

COs	POs												PSOs		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
1	3	3	2	2	2	-	-	-	1	-	1	1	3	2	1
2	3	3	2	2	2	-	-	-	1	-	1	1	3	2	1
3	3	3	2	2	2	-	-	-	1	-	1	1	3	2	1
4	3	3	2	2	2	-	-	-	1	-	1	1	3	2	1
5	3	3	2	2	2	-	-	-	1	-	1	1	3	2	1
6	3	3	2	2	2	-	-	-	1	-	1	1	3	2	1
Avg.	3	3	2	2	2	-	-	-	1	-	1	1	3	2	1

1 - Low, 2 - medium, 3 - high, ‘-“- no correlation

24ES102 ENGINEERING GRAPHICS

Offered by MECH (Common to AI&DS, CSE, CSE(AIML), CSBS & IT branches) (I Sem.)

L	T	P	C
2	0	4	4

COURSE OBJECTIVES

- To Understand the principles of drawing engineering curves.
- To Understand the projection of points, lines, plane surfaces and freehand sketching of simple objects.
- To Apply the methods of orthographic projection to solids.
- To Apply the methods of sections of solids.
- To Apply the skills to develop the surfaces of solids.
- To Analyze the methods of isometric projection for simple solids.

CONCEPTS AND CONVENTIONS (Not for Examination)

Importance of graphics in engineering applications – Use of drafting instruments - BIS conventions and specifications - Size, layout and folding of drawing sheets - Lettering and dimensioning.

UNIT I PLANE CURVES AND FREE HAND SKETCHING 6+9

Basic Geometrical constructions, Curves used in engineering practices: Conics - Construction of ellipse, parabola and hyperbola by eccentricity method - Drawing of tangents and normal to the above curves.
Visualization concepts and Free Hand sketching: Visualization principles - Representation of Three Dimensional objects - Layout of views- Freehand sketching of multiple views from pictorial views of objects.

UNIT II PROJECTION OF POINTS, LINES AND PLANE SURFACE 6+9

Orthographic projection – principles – Principal planes – First angle projection -projection of points. Projection of straight lines (only First angle projections) inclined to both the principal planes – Determination of true lengths and true inclinations by rotating line method.
Projection of planes (polygonal and circular surfaces) inclined to one principal plane. Projection of planes inclined to one principal plane using CAD Software (Not for examination).

UNIT III PROJECTION OF SOLIDS 6+9

Projection of simple solids like prisms, pyramids, cylinder, cone when the axis is inclined to one of the principal planes and parallel to the other by rotating object method.
Projection of simple solids like prisms and pyramids when the axis is inclined to one of the principal planes and parallel to the other using CAD Software (Not for examination).

UNIT IV PROJECTION OF SECTIONED SOLIDS AND DEVELOPMENT OF SURFACES 6+9

Sectioning of above solids in simple vertical position when the cutting plane is inclined to HP and perpendicular to VP—obtaining true shape of section.
Development of lateral surfaces of simple and sectioned solids when the cutting plane is inclined to HP and perpendicular to VP—Prisms, pyramids cylinders and cones.
Sectional view of a cone and cylinder in simple vertical position when the cutting plane is inclined to HP and perpendicular to VP using CAD Software (Not for examination).

UNIT V ISOMETRIC VIEWS AND PROJECTIONS 6+9

Principles of isometric view-Isometric view of simple solids and truncated solids-Prisms, pyramids, cylinders, cones in simple vertical positions.
Principles of isometric projection –isometric scale-Isometric projection of simple solids and truncated solids-Prisms, pyramids, cylinders, cones in simple vertical positions.

TOTAL: 30+45 PERIODS

TEXT BOOKS

1. Venugopal K and Prabhu Raja V, "Engineering Graphics + AutoCAD", New Age International (P) Limited, 2022.
2. Natrajan K.V., "A Text Book of Engineering Graphics", Dhanalakshmi Publishers, Chennai, 2018.

REFERENCES

1. Bhatt N.D. and Panchal V.M., "Engineering Drawing", Charotar Publishing House, 2023.
2. Parthasarathy, N.S. and VelaMurali, "Engineering Drawing", Oxford University Press, 2015.
3. Gopalakrishna K.R., "Engineering Drawing (Vol. I & II combined)", Subhas Publications, Bangalore, 2017.
4. Shah M.B., and Rana B.C., "Engineering Drawing", Pearson Education India, 2009.

Publication of Bureau of Indian Standards:

1. IS 10711—2001: Technical products Documentation—Size and layout of drawing sheets.
2. IS9609 (Parts0&1)—2001: Technical products Documentation—Lettering.
3. IS 10714(Part20)—2001&SP46—2003: Lines for technical drawings.
4. IS11669—1986&SP46—2003: Dimensioning of Technical Drawings.
5. IS 15021(Parts1to4)—2001: Technical drawings—Projection Methods.

Special points applicable to University Examinations on Engineering Graphics:

1. There will be five questions, each of either-or type covering all units of the syllabus.
2. All questions will carry equal marks of 20 each making a total of 100.
3. The answer paper shall consist of drawing sheets of A3 size only. The students will be permitted to use appropriate scale to fit solution with in A3 size.
4. The examination will be conducted in appropriate sessions on the same day.

COURSE OUTCOMES

After the completion of the course, the students will be able to

CO1: Understand the construction of conic curves and orthographic projections of simple solids.

CO2: Understand and solve practical problems involving the projection of points, lines, planes and freehand sketching of simple objects.

CO3: Apply the techniques for the projection of simple solids.

CO4: Apply the concepts of projection and sectioning of solids.

CO5: Apply the methods to draw sections and development of simple solids.

CO6: Analyze and draw the isometric views and projections of simple solids.

CO's-PO's & PSO's MAPPING

COs	POs												PSOs		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
1	3	1	2	-	2	-	-	-	-	3	-	2	2	2	-
2	3	1	2	-	2	-	-	-	-	3	-	2	2	2	-
3	3	1	2	-	2	-	-	-	-	3	-	2	2	2	-
4	3	1	2	-	2	-	-	-	-	3	-	2	2	2	-
5	3	1	2	-	2	-	-	-	-	3	-	2	2	2	-
6	3	1	2	-	2	-	-	-	-	3	-	2	2	2	-
Avg.	3	1	2	-	2	-	-	-	-	3	-	2	2	2	-

1 - Low, 2 - medium, 3 - high, '-"- no correlation

24TA101 HERITAGE OF TAMILS

Offered by Tamil (Common to ALL branches) (I Sem.)

L	T	P	C
1	0	0	1

UNIT I LANGUAGE AND LITERATURE **3**

Language Families in India-Dravidian Languages - Tamil as a Classical Language - Management Principles in Thirukural - Tamil Epics and Impact of Buddhism & Jainism in Tamil Land - Bakthi Literature Azhwars and Nayanmars - Forms of minor Poetry - Development of Modern literature in Tamil – Contribution of Bharathiyar and Bharathidhasan.

UNIT II HERITAGE- ROCK ART PAINTINGS TO MODERN ART- SCULPTURE **3**

Hero stone to modern sculpture - Bronze icons - Tribes and their handicrafts - Art of temple car making - Massive Terracotta sculptures, Village deities, Thiruvalluvar Statue at Kanyakumari, Making of musical instruments - Mridhangam, Parai, Veenai, Yazh and Nadhaswaram - Role of Temples in Social and Economic Life of Tamils.

UNIT III FOLK AND MARTIAL ARTS **3**

Therukoothu, Karagattam, Villu Pattu, Kaniyan Koothu, Oyillattam, Leather puppetry, Silambattam, Valari, Tiger dance - Sports and Games of Tamils.

UNIT IV THINAI CONCEPT OF TAMILS **3**

Flora and Fauna of Tamils & Aham and Puram Concept from Tholkappiyam and Sangam Literature - Aram Concept of Tamils - Education and Literacy during Sangam Age - Ancient Cities and Ports of Sangam Age - Export and Import during Sangam Age - Overseas Conquest of Cholas.

UNIT V CONTRIBUTION OF TAMILS TO INDIAN NATIONAL MOVEMENT AND INDIAN CULTURE **3**

Contribution of Tamils to Indian Freedom Struggle - The Cultural Influence of Tamils over the other parts of India – Self-Respect Movement - Role of Siddha Medicine in Indigenous Systems of Medicine – Inscriptions& Manuscripts – Print History of Tamil Books.

TOTAL: 15 PERIODS

24TA101 தமிழர் மரபு

Offered by Tamil (Common to ALL branches) (I Sem.)

L	T	P	C
1	0	0	1

அலகு I மொழி மற்றும் இலக்கியம் **3**

இந்திய மொழிக் குடும்பங்கள் - திராவிட மொழிகள் - தமிழ் ஒரு செம்மொழி - தமிழ் செவ்விலக்கியங்கள் - சங்க இலக்கியத்தின் சமயச் சார்பற்ற தன்மை - சங்க இலக்கியத்தில் பகிர்தல் அறம் - திருக்குறளில் மேலாண்மைக் கருத்துக்கள் - தமிழ்க் காப்பியங்கள், தமிழகத்தில் சமண பௌத்த சமயங்களின் தாக்கம் - பக்தி இலக்கியம், ஆழ்வார்கள் மற்றும் நாயன்மார்கள் - சிற்றிலக்கியங்கள் - தமிழில் நவீன இலக்கியத்தின் வளர்ச்சி - தமிழ் இலக்கிய வளர்ச்சியில் பாரதியார் மற்றும் பாரதிதாசன் ஆகியோரின் பங்களிப்பு.

அலகு II மரபு-பாறை ஓவியங்கள் முதல் நவீன ஓவியங்கள் வரை - சிற்பக்கலை **3**

நடுகல் முதல் நவீன சிற்பங்கள் வரை - ஐம்பொன் சிலைகள் - பழங்குடியினர்மற்றும் அவர்கள் தயாரிக்கும் கைவினைப் பொருட்கள், பொம்மைகள் - தேர்செய்யும் கலை - சுடுமண் சிற்பங்கள் - நாட்டுப்புறத் தெய்வங்கள் -குமரிமுனையில் திருவள்ளுவர் சிலை - இசைக் கருவிகள் - மிருதங்கம், பறை, வீணை, யாழ், நாதஸ்வரம் - தமிழர்களின் சமூக பொருளாதார வாழ்வில் கோவில்களின் பங்கு. நடுகல் முதல் நவீன சிற்பங்கள் வரை -

ஐம்பொன் சிலைகள்- பழங்குடியினர் மற்றும் அவர்கள் தயாரிக்கும் கைவினைப் பொருட்கள், பொம்மைகள் - தேர்செய்யும் கலை - சுடுமண் சிற்பங்கள் - நாட்டுப்புறத் தெய்வங்கள் - குமரிமுனையில் திருவள்ளுவர் சிலை - இசைக் கருவிகள் - மிருதங்கம், பறை, வீணை, யாழ், நாதஸ்வரம் - தமிழர்களின் சமூக பொருளாதார வாழ்வில் கோவில்களின் பங்கு.

அலகு III நாட்டுப்புறக் கலைகள் மற்றும் வீர விளையாட்டுகள் 3

தெருக்கூத்து, கரகாட்டம், வில்லுப்பாட்டு, கணியான் கூத்து, ஒயிலாட்டம், தோல்பாவைக் கூத்து, சிலம்பாட்டம், வளரி, புலியாட்டம், தமிழர்களின்விளையாட்டுகள்.

அலகு IV தமிழர்களின் திணைக் கோட்பாடுகள் 3

தமிழகத்தின் தாவரங்களும், விலங்குகளும் - தொல்காப்பியம் மற்றும் சங்க இலக்கியத்தில் அகம் மற்றும் புறக் கோட்பாடுகள் - தமிழர்கள் போற்றிய அறக்கோட்பாடு - சங்ககாலத்தில் தமிழகத்தில் எழுத்தறிவும், கல்வியும் -சங்ககால நகரங்களும் துறை முகங்களும் - சங்ககாலத்தில் ஏற்றுமதி மற்றும் இறக்குமதி - கடல்கடந்த நாடுகளில் சோழர்களின் வெற்றி.

அலகு V இந்திய தேசிய இயக்கம் மற்றும் இந்திய பண்பாட்டிற்குத்தமிழர்களின் பங்களிப்பு 3

இந்திய விடுதலைப்போரில் தமிழர்களின் பங்கு - இந்தியாவின்பிறப்பகுதிகளில் தமிழ்ப் பண்பாட்டின் தாக்கம் - சுயமரியாதை இயக்கம் -இந்திய மருத்துவத்தில், சித்த மருத்துவத்தின் பங்கு - கல்வெட்டுகள், கையெழுத்துப்படிகள் - தமிழ்ப் புத்தகங்களின் அச்சு வரலாறு.

TOTAL: 15 PERIODS

TEXT-CUM-REFERENCE BOOKS:

1. தமிழக வரலாறு - மக்களும் பண்பாடும் - கே.கே. பிள்ளை (வெளியீடு: தமிழ்நாடு பாடநூல் மற்றும் கல்வியியல் பணிகள் கழகம்).
2. கணினித் தமிழ் - முனைவர் இல. சுந்தரம். (விகடன் பிரசுரம்).
3. கீழடி - வைகை நதிக்கரையில் சங்ககால நகர நாகரிகம் (தொல்லியல் துறை வெளியீடு).
4. பொருநை - ஆற்றங்கரை நாகரிகம். (தொல்லியல் துறை வெளியீடு).
5. Social Life of Tamils (Dr.K.K.Pillay) A joint publication of TNTB & ESC and RMRL - (in print)
6. Social Life of the Tamils - The Classical Period (Dr.S.Singaravelu) (Published by: International Institute of Tamil Studies.
7. Historical Heritage of the Tamils (Dr.S.V.Subatamanian, Dr.K.D. Thirunavukkarasu) (Published by: International Institute of Tamil Studies).
8. The Contributions of the Tamils to Indian Culture (Dr.M.Valarmathi) (Published by: International Institute of Tamil Studies.)
9. Keeladi - 'Sangam City Civilization on the banks of river Vaigai' (Jointly Published by: Department of Archaeology & Tamil Nadu Text Book and Educational Services Corporation, Tamil Nadu)
10. Studies in the History of India with Special Reference to Tamil Nadu (Dr.K.K.Pillay) (Published by: The Author)
11. Porunai Civilization (Jointly Published by: Department of Archaeology & Tamil Nadu Text Book and Educational Services Corporation, Tamil Nadu)
12. Journey of Civilization Indus to Vaigai (R.Balakrishnan) (Published by: RMRL) - Reference Book.

24PH111 PHYSICS AND CHEMISTRY LABORATORY

Offered by Phy. & Che. (Common to ALL branches) (I Sem.)

L	T	P	C
0	0	4	2

PHYSICS LABORATORY (Any Six Experiments)

COURSE OBJECTIVES

- To learn various kinds of physics laboratory equipment.
- To learn problem solving skills related to physics principles and interpretation of experimental data.
- To measure experimental values and learn techniques for active participation in all laboratory exercises.

LIST OF EXPERIMENTS

1. Torsional pendulum–Determination of rigidity modulus of wire and moment of inertia of regular and irregular objects.
2. Simple harmonic oscillations of cantilever.
3. Non-uniform bending–Determination of Young's modulus.
4. Uniform bending–Determination of Young's modulus.
5. Laser–Determination of wave length of laser using grating.
6. Air wedge – Determination of thickness of a thin sheet/wire.
7. a) Optical fiber– Determination of numerical aperture and acceptance angle.
   b) Compact disc–Determination of width of a groove using laser.
8. Acoustic grating–Determination of velocity of ultrasonic waves in liquid.
9. Ultrasonic interferometer–Determination of velocity of sound and compressibility of liquid.
10. Post office box–Determination of band gap of a semiconductor.

TOTAL: 30 PERIODS

CHEMISTRY LABORATORY (Any Seven Experiments) COURSE

OBJECTIVES

- Understand and apply the basic techniques involved in quantitative analysis.
- Familiarize sophisticated analytical equipment.
- Application of the knowledge gained in theory course.

LIST OF EXPERIMENTS

1. Estimation of total, permanent and temporary hardness of water by EDTA method.
2. Estimation of nickel content in steel by EDTA method.
3. Estimation of types and amount of alkalinity of water sample.
4. Estimation of iron content of the water sample by spectrophotometry.
5. Determination of the strength of hydrochloric acid by PHmetry.
6. Determination of the strength of strong acid by conductometry.
7. Determination of the strength of mixture of strong and weak acids using conductometry.
8. Estimation of sodium in water by flame photometry.
9. Estimation of Fe^{2+}ofthe given solution using potentio metric titration.
10. Determination of rate of corrosion of mild steel in acidic medium.
11. Conduct ometric titration of barium chloride against sodium sulphate (precipitation titration).

TOTAL: 30 PERIODS

COURSE OUTCOMES

After the completion of the course, the students will be able to

CO1: Understand the functioning of various physics laboratory equipment.

CO2: Solve various problems by using principles of physics and experimental data.

CO3: Access, process and analyze scientific information individually and collaboratively.

CO4: Analyse various water quality parameters like hardness, alkalinity etc., in water sample.

CO5: Acquire practical skills by using instruments like conductivity meter, pHmeter, Potentiometer, spectrophotometer etc.,

CO6: Finding the strength and amount of metalions in various alloys.

CO's-PO's & PSO's MAPPING

COs	POs												PSOs		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
1	3	2	2	1	2	-	-	-	-	-	-	-	-	-	3
2	3	2	2	1	2	-	-	-	-	-	-	-	-	-	3
3	3	2	2	1	2	-	-	-	-	-	-	-	-	-	3
4	3	2	2	1	-	2	-	-	-	-	-	-	-	-	3
5	3	2	2	1	2	2	-	-	-	-	-	-	-	-	3
6	3	2	2	1	2	2	-	-	-	-	-	-	-	-	3
Avg.	3	2	2	1	2	2	-	-	-	-	-	-	-	-	3

1 - Low, 2 - medium, 3 - high, '-"- no correlation

24ES111 PYTHON PROGRAMMING LABORATORY

Offered by CSE (Common to ALL branches) (I Sem.)

L	T	P	C
0	0	4	2

COURSE OBJECTIVES

- To understand the problem-solving approaches.
- To learn the basic programming constructs in Python.
- To practice various computing strategies for Python-based solutions to real world problems.
- To use Python data structures- lists, tuples, dictionaries.
- To do input/output with files in Python.

LIST OF EXPERIMENTS

Note: The examples suggested in each experiment are only indicative. The lab instructor is expected to design other problems on similar lines. The Examinations Hall not be restricted to the sample experiments listed here.

1. Identification and solving of simple real life or scientific or technical problems, and developing flow charts for the same (electricity billing, retail shop billing, sine series, weight of a motorbike, weight of a steel bar, compute electrical current in three phase AC circuit, etc.)
2. Python programming using simple statements and expressions (exchange the values of two variables, circulate the values of n variables, distance between two points).
3. Scientific problems using Conditionals and Iterative loops (number series, number patterns, pyramid pattern)
4. Implementing real-time/technical applications using Lists, Tuple (items present in a library/ component of a car/materials required for construction of a building –operations of list &tuple)
5. Implementing real time/technical applications using Sets, Dictionaries (components of an automobile, elements of a civil structure, etc. –operations of sets & dictionaries)
6. Implementing programs using Functions (factorial, largest number in a list, area of shape)
7. Implementing programs using Strings (rever`se, palindrome, character count, replacing characters)
8. Implementing programs using written modules and Python Standard Libraries (pandas, numpy, mat plot lib, scipy)
9. Implementing real-time/technical applications using File handling (copy from one file to another, word count, longest word)
10. Implementing real-time/technical applications using Exception handling (divide by zero error, voter's age validity, student mark range validation)
11. Developing a game activity using Pygame like bouncing ball, car race etc.

TOTAL: 60 PERIODS

COURSE OUTCOMES

After the completion of the course, the students will be able to

CO1: Develop algorithmic solutions to simple computational problems.

CO2: Develop and execute simple Python programs.

CO3: Implement programs in Python using conditionals and loops for solving problems.

CO4: Deploy functions to decompose a Python program.

CO5: Process compound data using Python data structures.

CO6: Utilize Python packages in developing software applications

TEXT BOOKS

1. S.Sridhar, J. Indumathi, V.M. Hariharan, "Python Programming", Pearson Education, 2023.

REFERENCES

1. Paul Deitel and Harvey Deitel, "Python for Programmers", Pearson Education, 1st Edition, 2021.
2. G Venkatesh and Madhavan Mukund, "Computational Thinking: A Primer for Programmers and Data Scientists", 1st Edition, Notion Press, 2021.
3. John V Guttag, "Introduction to Computation and Programming Using Python: With Applications to Computational Modeling and Understanding Data'', 3rd Edition, MIT Press, 2021.
4. R. Nageswara Rao, "Core Python Programming", 3rd Edition, Dreamtech Press, 2021.
5. Allen B. Downey, "Think Python: How to Think Like a Computer Scientist", 2nd Edition, O'Reilly, 2016.
6. Ashok Namdev Kamthane and Amit Ashok Kamthane, "Python Programming", McGraw Hill Education (India) Private Limited, 2018.

CO's-PO's & PSO's MAPPING

COs	POs												PSOs		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
1	3	3	2	2	2	-	-	-	2	-	1	1	3	2	1
2	3	3	2	2	2	-	-	-	2	-	1	1	3	2	1
3	3	3	2	2	2	-	-	-	2	-	1	1	3	2	1
4	3	3	2	2	2	-	-	-	2	-	1	1	3	2	1
5	3	3	2	2	2	-	-	-	2	-	1	1	3	2	1
6	3	3	2	2	2	-	-	-	2	-	1	1	3	2	1
Avg.	3	3	2	2	2	-	-	-	2	-	1	1	3	2	1

1 - Low, 2 - medium, 3 - high, '-"- no correlation

24EM111 **COMMUNICATION LABORATORY I**
Offered by English (Common to ALL branches) (I Sem.)

L	T	P	C
0	0	2	1

COURSE OBJECTIVES

- To improve the communicative competence of learners
- To help learners use language effectively in academic/ work contexts
- To develop various listening strategies to comprehend various types of audio materials like lectures, discussions, videos etc.
- To build on students' English language skills by engaging them in listening, speaking and grammar learning activities those are relevant to authentic contexts
- To use language efficiently in expressing their opinions via various media
- To support students in improving both their speaking and listening abilities

UNIT I SELF-INTRODUCTION **6**

Listening - General information-specific details, Listening to lectures, class room lectures and monologues, Telephone/Email etiquettes.

Speaking - Self-Introduction; Introducing a friend; Conversation - Telephone - Making polite requests, making polite offers, replying to polite requests and offers –Understanding basic instructions (filling out a bank application for example)

UNIT II NARRATION **6**

Listening - Listening to podcasts, anecdotes, TED Talks / stories / event narration; documentaries and interviews with celebrities.

Speaking - Narrating personal experiences / events - Talking about current and temporary situations.

UNIT III PROCESS AND PRODUCT DESCRIPTION **6**

Listening - Listening to Tales / Fables, Songs & Lyrics, movies & short film, News & Media.

Speaking - Picture description - Describing locations in workplaces - Giving instruction to use the product - Explaining uses and purposes –Presenting a product.

UNIT IV EXTEMPORE SPEECHES **6**

Listening - Listening to Video blogs; Listening to lectures and educational videos.
Speaking - Small talk, discussing and making plans - talking about tasks - talking about progress - talking about positions and directions of movement talking about travel preparations talking about transportation

UNIT V GROUP DISCUSSION **6**

Listening - Listening to debates/discussions; different viewpoints on an issue and panel discussions.
Speaking - Public speaking - Group discussion - Debates and Discussions

TOTAL: 30 PERIODS

COURSE OUTCOMES

After the completion of the course, the students will be able to

CO1: To listen and comprehend general as well as complex academic information
CO2: To observe and understand different points of view in a discussion
CO3: To speak fluently and accurately informal and informal communicative contexts
CO4: To describe products and processes and explain their uses and purposes clearly and accurately
CO5: To express their opinions effectively in both formal and informal discussions
CO6: To use different communicative functions effectively

CO's-PO's & PSO's MAPPING

COs	POs												PSOs		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
1	-	-	-	1	-	1	1	2	3	3	1	2	-	-	-
2	-	-	-	1	-	1	1	2	3	3	1	2	-	-	-
3	-	-	-	1	-	1	1	2	3	3	1	2	-	-	-
4	-	-	-	1	-	1	1	2	3	3	1	2	-	-	-
5	-	-	-	1	-	1	1	2	3	3	1	2	-	-	-
6	-	-	-	1	-	1	1	2	3	3	1	2	-	-	-
Avg.	-	-	-	1	-	1	1	2	3	3	1	2	-	-	-

1 - Low, 2 - medium, 3 - high, '-"- no correlation

24EN201 PROFESSIONAL ENGLISH II

Offered by English (Common to ALL branches) (II Sem.)

L	T	P	C
3	0	0	3

COURSE OBJECTIVES

- To engage learners in meaningful language activities to improve their reading and writing skills for formal and informal communication like Email/business letters.
- To help learners understand the strategies in reading, drafting societal communication.
- To apply various reading mechanics to comprehend sentences incoherent order and to enhance writing skill.
- To hone their grammatical skills and improve report writing skill.
- To demonstrate and understanding of job applications and interviews for internship and placements.
- To improve the verbal ability skill and communicative skill of the students.

UNIT I SENTENCE PATTERN 9

Reading - Skimming and scanning, Reading comprehension. **Writing -** Professional emails, Email etiquette, Writing business letters. **Grammar -** Mixed tenses, Sentence pattern, Construction of sentences; **Vocabulary -** Phrasal verbs.

UNIT II ART OF LETTER WRITING9

Reading - Speed reading-reading passages with time limit. **Writing -** Writing letter of complaints/responses to complaints, Letter to the Editor. **Grammar -** Active Passive voice transformations. **Vocabulary -** Idiomatic expressions.

UNIT III FORMAL COMMUNICATION 9

Reading - Reading comprehension and Extensive reading on short stories/novels. **Writing -** Jumbled sentences, Checklists, Minutes of meeting. **Grammar -** If conditional sentences, Numerical expressions. **Vocabulary -** Sentence completion –Cloze test.

UNIT IV REPORT WRITING 9

Reading - Reading job advertisements/job descriptions/profile of the company. **Writing -** Report writing - Industrial visit - Accident - Feasibility - Survey, Article writing. **Grammar -** Reported Speech, Modals. **Vocabulary -** Conjunctions-Use of prepositions.

UNIT V JOB APPLICATION AND RESUME WRITING 9

Reading - Critical reading and inferences. **Writing -** Job / Internship application – Cover letter & Resume /visual resume. **Grammar -** Error correction/Deduction–Articles–Single word substitution. **Vocabulary -** Misspelt words–Commonly confused words –Homophones and Homonyms.

TOTAL: 45 PERIODS

TEXT BOOKS

1. Department of English, “English for Engineers & Technologists”, Anna University, Orient Black swan Private Ltd, 2020.
2. Dr. Veena Selvam, Dr. Sujatha Priyadarshini, Dr. Deepa Mary Francis, Dr. KN.Shoba, and Dr. Lourdes Joevani, “English for Science & Technology”, Cambridge University Press, 2021.

REFERENCES

1. Meenakshi Raman & Sangeeta Sharma, “Technical Communication – Principles and Practices”, Oxford Univ. Press, New Delhi. 2016.
2. V.N.Arora and LaxmiChandraed, “Improve Your Writing ”Oxford University Press, NewDelhi,2001.
3. Dr.V.Chellammal, “Learning to Communicate, ”Allied Publishing House, NewDelhi, 2003.
4. R.C.Sharma & KrishnaMohan, “Business Correspondence and Report Writing ”,Tata Mc Graw Hill & Co. Ltd., NewDelhi. 2001.
5. Krishna Mohan, Meera Bannerji, “Developing Communication Skills,” Macmillan India Ltd. New Delhi, 1990.

COURSE OUTCOMES

After the completion of the course, the students will be able to

CO1: To write cohesively, coherently and flawlessly avoiding grammatical errors using appropriate communicative strategies

CO2: To identify and report cause and effects in events, industrial processes through technical texts

CO3: To analyze problems in order to arrive at feasible solutions and communicate them in the written format.

CO4: To present their ideas and opinions in a planned and logical manner

CO5: To prepare various reports

CO6: To draft effective resumes in the context of job search

CO's-PO's & PSO's MAPPING

COs	POs												PSOs		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
1	-	-	-	-	-	-	-	2	3	3	1	1	-	-	-
2	-	-	-	-	-	-	-	2	3	3	1	1	-	-	-
3	-	-	-	-	-	-	-	2	3	3	1	1	-	-	-
4	-	-	-	-	-	-	-	2	3	3	1	1	-	-	-
5	-	-	-	-	-	-	-	2	3	3	1	1	-	-	-
6	-	-	-	-	-	-	-	2	3	3	1	1	-	-	-
Avg.	-	-	-	-	-	-	-	2	3	3	1	1	-	-	-

1 - Low, 2 - medium, 3 - high, '-"- no correlation

24MA201 **ENGINEERING MATHEMATICS II**

Offered by Maths (Common to ALL branches) (II Sem.)

L	T	P	C
3	1	0	4

COURSE OBJECTIVES

- To gain knowledge in differential equations essential for engineering problems.
- To apply complex analysis and Laplace transforms to solve engineering issues.
- To familiarize fundamental concepts properties of Z-transforms and involve the solutions of differential equations.
- To introduce and solve algebraic and transcendental equations using direct methods.
- To solve simultaneous linear equations with iterative methods and address eigenvalue problems.
- To apply numerical techniques for interpolation, differentiation, and integration in engineering.

UNIT I ORDINARY DIFFERENTIAL EQUATIONS **9+3**

Higher order linear differential equations with constant coefficients–Method of variation of parameters – Homogenous equation of Euler's and Legendre's type–System of simultaneous linear first order differential equations with constant coefficients.

UNIT II LAPLACE TRANSFORMS **9+3**

Existence conditions – Transforms of elementary functions – Transform of unit step function and unit impulse function–Basic properties–Shifting theorems–Transforms of derivatives and integrals – Initial and final value theorems (Statements only)– Inverse transforms – Partial fraction and convolution theorem – Transform of periodic functions – Application to solution of linear second order ordinary differential equations with constant coefficients.

UNIT III Z-TRANSFORMS AND DIFFERENCE EQUATIONS **9+3**

Z-transforms–Elementary properties–Convergence of Z-transforms–Initial and final value theorems– Inverse Z-transforms using partial fraction and convolution theorem–Formation of difference equations –Solution of difference equations using Z-transforms.

UNIT IV SOLUTION OF EQUATIONS AND EIGEN VALUE PROBLEMS **9+3**

Solution of algebraic and transcendental equations– Fixed point iteration method–Newton Raphson method–Solution of linear system of equations – Gauss elimination method – Gauss Jordan method – Iterative methods of Gauss Jacobi and Gauss Seidel – Eigen values of a matrix by Power method.

UNIT V INTERPOLATION, NUMERICAL DIFFERENTIATION AND INTEGRATION **9+3**

Lagrange's and Newton's divided difference interpolations–Newton's forward and backward difference interpolation – Approximation of derivates using interpolation polynomials – Numerical single and double integrations using Trapezoidal and Simpson's1/3rules.

TOTAL: 45+15 PERIODS

TEXT BOOKS

1. Grewal, B.S. and Grewal, J.S., "Numerical Methods in Engineering and Science", Khanna Publishers, NewDelhi, 11th Edition, 2017.
2. Erwin Kreyszig, "Advanced Engineering Mathematics", John Wiley & Sons,10th Edition, 2017.

REFERENCES

1. Larry C Andrews, Bhimsen K Shivamoggi, "Integral Transforms for Engineers", Prentice Hall of India, 2005.
2. Burden R.L. and Faires J.D., "Numerical Analysis",CengageLearning,10thEdition,2016.
3. Gerald, C.F., and Wheatley, P.O., "Applied Numerical Analysis", Pearson Education, Asia, NewDelhi, 7th Edition, 2007.
4. Bali, N.P., and Manish Goyal, "A Textbook of Engineering Mathematics", Laxmi Publications Pvt. Ltd, 11th Edition,2022.

5. Narayanan, S. and Manigavachagom Pillay, T.K., Dr.Ramanaiah,G., “Advanced Mathematics for Engineering Students– Volume II and III”, Ananda Book Depot, 2019.
6. Ramana B.V., “Higher Engineering Mathematics”, McGraw Hill Education Pvt. Ltd, NewDelhi,2018.

COURSE OUTCOMES

After the completion of the course, the students will be able to

CO1: Understand differential equations commonly used in engineering applications.

CO2: Apply Laplace transform techniques to solve physical engineering problems.

CO3: Utilize Z-transform techniques to address real-time discrete problems.

CO4: Solve algebraic, transcendental, and simultaneous equations using direct methods numerically.

CO5: Compute solutions for simultaneous equations using it erative methods and determine approximate eigen values.

CO6: Interpret numerical techniques for interpolation, differentiation, and integration in engineering contexts.

CO’s-PO’s & PSO’s MAPPING

COs	POs												PSOs		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
1	3	3	2	2	-	-	-	-	2	-	-	2	-	-	-
2	3	3	2	2	-	-	-	-	2	-	-	2	-	-	-
3	3	3	2	2	-	-	-	-	2	-	-	2	-	-	-
4	3	3	2	2	-	-	-	-	2	-	-	2	-	-	-
5	3	3	2	2	-	-	-	-	2	-	-	2	-	-	-
6	3	3	2	2	-	-	-	-	2	-	-	2	-	-	-
Avg.	3	3	2	2	-	-	-	-	2	-	-	2	-	-	-

1 - Low, 2 - medium, 3 - high, ‘-“- no correlation

24PH201 **PHYSICS FOR INFORMATION SCIENCE**

Offered by Physics (Common to AIDS, CSE, CSE(AIML), CSBS & IT branches) (II Sem.)

L	T	P	C
3	0	0	3

COURSE OBJECTIVES

- To understand the fundamental concepts of electrical properties of materials.
- To gain knowledge on semiconductors and their applications.
- To get knowledge on the magnetic properties of materials and their devices.
- To establish a sound grasp of knowledge on different optical properties of materials, optical devices and applications.
- To learn the importance of nanostructures, quantum confinement and applications in nano devices.
- To know the fundamental concepts of quantum computing.

UNIT I ELECTRICAL PROPERTIES OF MATERIALS **9**

Classical free electron theory – Expression for electrical conductivity and thermal conductivity – Wiedemann-Franz law – Success and failures of classical free electron theory – Quantum free electron theory – Degenerate states – Fermi-Dirac statistics – Density of energy states – Electron in periodic potential – Tight binding approximation – Electron effective mass – concept of hole.

UNIT II SEMICONDUCTOR PHYSICS **9**

Intrinsic semiconductors – Energy band diagram – Direct and indirect band gap semiconductors –Carrier concentration in intrinsic semiconductors – Extrinsic semiconductors – Carrier concentration in N-type & P-type semiconductors – Variation of carrier concentration with temperature – Variation of Fermi level with temperature and impurity concentration – Carrier transport in semiconductor: random motion, drift, mobility and diffusion – Hall effect – Ohmic contacts – Schottky diode.

UNIT III MAGNETIC PROPERTIES OF MATERIALS **9**

Magnetic dipole moment – Atomic magnetic moments – Magnetic permeability and susceptibility – Magnetic material classification: diamagnetism – paramagnetism – ferromagnetism – antiferromagnetism – ferrimagnetism – Saturation magnetization and curie temperature – M versus H behaviour – Hard and soft magnetic materials – GMR sensor.

UNIT IV OPTICAL PROPERTIES OF MATERIALS **9**

Classification of optical materials – Carrier generation and recombination processes – Absorption, emission and scattering of light in metals, insulators and semiconductors – Photodiode – Solar cell – Light Emitting Diode (LED) – Organic LED – Laser diode – Optical data storage techniques.

UNIT V NANO DEVICES AND QUANTUM COMPUTING **9**

Quantum confinement – Quantum structures: quantum well, quantum wire and quantum dot – Band gap of nanomaterials – Tunneling – Single electron phenomena – Coulomb blockade – Resonant tunneling diode – Single electron transistor – Quantum computing – classical bits – quantum bits or qubits – single qubits – multiple qubits– Bloch sphere – quantum gates – Advantage of quantum computing over classical computing.

TOTAL: 45 PERIODS

COURSE OUTCOMES

At the end of the course, the students will be able to

CO1: Gain the fundamental knowledge on electrical properties of materials.
CO2: Have knowledge on semiconductors and their applications on various devices.
CO3: Get knowledge on magnetic properties of materials and their applications in data storage.
CO4: Have the necessary understanding on the functioning of optical materials for optoelectronic devices.
CO5: Obtain the basic knowledge of quantum structures and the applications of nano devices.
CO6: Know the fundamental concepts of quantum computing.

TEXTBOOKS

1. Donald Neamen, "Semiconductor Physics and Devices", Mc Graw-Hill, 2021.
2. S.O. Kasap, " Principles of Electronic Materials and Devices", Mc Graw-Hill Education (Indian Edition), 2020.
3. K. Parag Lala, " Quantum Computing: A Beginner's Introduction", Mc Graw-Hill Education (Indian Edition), 2020.

REFERENCE BOOKS

1. Charles Kittel, "Introduction to Solid State Physics", Wiley India Edition, 2019.
2. B. Rogers, J. Adams and S. Pennathur, "Nanotechnology: Understanding Small Systems", CRC Press, 2014.
3. Y.B. Band and Y. Avishai, "Quantum Mechanics with Applications to Nanotechnology and Information Science", Academic Press, 2013.
4. G.W. Hanson, "Fundamentals of Nanoelectronics", Pearson Education (Indian Edition) 2009.
5. V.V. Mitin, V.A. Kochelap and M.A.Stroscio, "Introduction to Nano electronics", Cambridge University Press, 2008.

CO's-PO's & PSO's MAPPING

COs	POs												PSOs		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
1	3	1	2	-	-	-	-	-	-	-	-	-	-	-	-
2	3	1	2	-	-	-	-	-	-	-	-	-	-	-	-
3	3	1	2	2	2	1	2	-	-	-	-	-	-	-	-
4	3	1	2	2	2	1	2	-	-	-	-	-	-	-	-
5	3	1	2	2	2	1	2	-	-	-	-	2	-	-	-
6	3	1	2	2	2	1	2	-	-	-	-	2	-	-	-
Avg.	3	1	2	2	2	1	2	-	-	-	-	2	-	-	-

1 - low, 2 - medium, 3 - high, '-'- no correlation

24CS201 PROGRAMMING IN C

Offered by CSE (Common to AIDS, CSE, CSE(AIML), CSBS & IT branches) (II Sem.)

L	T	P	C
3	0	0	3

COURSE OBJECTIVES

- To understand the structure of C Language
- To develop C Programs using basic programming constructs
- To develop C programs using arrays and strings
- To develop modular applications in C using functions
- To develop applications in C using pointers and structures
- To understand the file handling in C

UNIT I INTRODUCTION TO C PROGRAMMING 9

Basic structure of C program, Constants, Variable and Data Types, Input and Output Operations, Operators and Expressions, Evaluation of Expressions, Type Conversions in Expressions, Operator Precedence and Associativity.

UNIT II CONTROL STRUCTURES 9

Decision Making with IF, IF-ELSE, Nested IF Statement, Switch statement, Conditional Operator, while Statement, do statement, for statement, Nested for statements, Unconditional statements.

UNIT III ARRAYS AND STRINGS 9

One-dimensional Arrays, Two-dimensional Arrays, Character Arrays and Strings, Arithmetic Operations on Characters, String-handling Functions.

UNIT IV FUNCTIONS AND POINTERS 9

Definition of Functions, User-defined functions, Types of functions and return values, Call by value, Recursion. Pointers: Declaring pointer variables, Initialization of pointer variables, Accessing pointer variables, Pointer expressions, Call by reference and Passing arrays to functions, Storage Classes.

UNIT V STRUCTURES AND FILE MANAGEMENT 9

Structures: Defining a structure, Declaring structure variables, Accessing structure members, Structure initialization, Array of structures, Pointer to structures, Union. File Management: Defining and opening a file, closing a file, Input/output and Error Handling on Files.

TOTAL: 45 PERIODS

COURSE OUTCOMES:

Upon completion of the course, the students will be able to

CO1: Demonstrate knowledge on C Programming constructs
CO2: Develop simple applications in C using appropriate control structures
CO3: Design and implement applications using arrays and strings
CO4: Design user defined functions and apply concept of recursion to solve problems
CO5: Develop applications in C using structures and pointers
CO6: Design applications using sequential and random access file processing

TEXT BOOKS:

1. E. Balaguruswamy, "Programming in ANSI C", 8th Edition, 2019, McGraw Hill Education, ISBN: 978-93-5316-513-0.
2. Yashavant P. Kanetkar, "Let Us C", 19th Edition, 2022, BPB Publications, ISBN 13: 978-9355512765.

REFERENCE BOOKS:

1. Kernighan B.W and Dennis M. Ritchie, "The C Programming Language", 2nd Edition, 2015, Pearson Education India, ISBN: 978-93-3254-944-9.
2. Reema Thareja, "Programming in C", Oxford university press, Second Edition, 2016, ISBN: 9780199456147
3. Anita Goel, Ajay Mittal, "Computer Fundamentals and Programming in C", Pearson Education India , 2016. ISBN: 9789332579200
4. Jacqueline A Jones and Keith Harrow, "Problem Solving with C", Pearson Education. ISBN: 978-93-325-3800-9.

CO's-PO's & PSO's MAPPING

COs	POs												PSOs		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
1	3	3	2	2	1	-	-	-	1	-	1	1	3	2	1
2	3	3	2	2	1	-	-	-	1	-	1	1	3	2	1
3	3	3	2	2	1	-	-	-	1	-	1	1	3	2	1
4	3	3	2	2	1	-	-	-	1	-	1	1	3	2	1
5	3	3	2	2	1	-	-	-	1	-	1	1	3	2	1
6	3	3	2	2	1	-	-	-	1	-	1	1	3	2	1
Avg.	**3**	**3**	**2**	**2**	**1**	-	-	-	**1**	-	**1**	**1**	**3**	**2**	**1**

1 - low, 2 - medium, 3 - high, '-'- no correlation

24TA201 TAMILS AND TECHNOLOGY

Offered by Tamil (Common to ALL branches) (II Sem.)

L	T	P	C
1	0	0	1

UNIT I WEAVING AND CERAMIC TECHNOLOGY 3

Weaving Industry during Sangam Age - Ceramic technology - Black and Red Ware Potteries (BRW) - Graffition Potteries.

UNIT II DESIGN AND CONSTRUCTION TECHNOLO 3

Designing and Structural construction House & Designs in house hold materials during Sangam Age - Building materials and Hero stones of Sangamage – Details of Stage Constructions in Silappathikaram - Sculptures and Temples of Mamallapuram - Great Temples of Cholas and other worship places - Temples of Nayaka Period - Type study (Madurai Meenakshi Temple) - Thirumalai Nayakar Mahal – ChettiNadu Houses, Indo - Saracenic architecture at Madras during British Period.

UNIT III MANUFACTURING TECHNOLOGY 3

Art of Ship Building - Metallurgical studies - Iron industry - Iron smelting, steel - Copper and gold - Coins assource of history – Minting of Coins - Beads making – industries Stone beads – Glass beads Terracotta beads -Shell beads/ bone beats - Archeological evidences - Gem stone types described in Silappathikaram.

UNIT IV AGRICULTURE AND IRRIGATION TECHNOLOGY 3

Dam, Tank, ponds, Sluice, Significance of Kumizhi Thoompu of Chola Period, Animal Husbandry - Wells designed for cattle use - Agriculture and Agro Processing - Knowledge of Sea - Fisheries – Pearl - Conche diving - Ancient Knowledge of Ocean - Knowledge Specific Society.

UNIT V SCIENTIFIC TAMIL & TAMIL COMPUTING 3

Development of Scientific Tamil - Tamil computing - Digitalization of Tamil Books - Development of Tamil Software - Tamil Virtual Academy - Tamil Digital Library - Online Tamil Dictionaries - Sorkuvai Project.

24TA201 தமிழரும் தொழில்நுட்பமும்

Offered by Tamil (Common to ALL branches) (II Sem.)

L	T	P	C
1	0	0	1

அலகு I நெசவு மற்றும் பானைத் தொழில்நுட்பம் 3

சங்க காலத்தில் நெசவுத் தொழில் - பானைத் தொழில்நுட்பம் - கருப்பு சிவப்பு பாண்டங்கள் - பாண்டங்களில் கீறல் குறியீடுகள்.

அலகு II வடிவமைப்பு மற்றும் கட்டிடத் தொழில்நுட்பம் 3

சங்க காலத்தில் வடிவமைப்பு மற்றும் கட்டுமானங்கள் & சங்க காலத்தில் வீட்டுப் பொருட்களில் வடிவமைப்பு- சங்க. காலத்தில் கட்டுமான பொருட்களும் நடுகல்லும் - சிலப்பதிகாரத்தில் மேடை அமைப்பு பற்றிய விவரங்கள் - மாமல்லபுரச் சிற்பங்களும், கோவில்களும் - சோழர் காலத்துப் பெருங்கோயில்கள் மற்றும் பிற வழிபாட்டுத் தலங்கள் - நாயக்கர் காலக் கோயில்கள் - மாதிரி கட்டமைப்புகள் பற்றி அறிதல், மதுரை மீனாட்சி அம்மன் ஆலயம் மற்றும் திருமலை நாயக்கர் மஹால் - செட்டிநாட்டு வீடுகள்- பிரிட்டிஷ் காலத்தில் சென்னையில் இந்தோ-சாரோசெனிக் கட்டிடக் கலை.

அலகு III உற்பத்தித் தொழில் நுட்பம் 3

கப்பல் கட்டும் கலை - உலோகவியல் - இரும்புத் தொழிற்சாலை - இரும்பை உருக்குதல் , எஃகு - வரலாற்றுச் சான்றுகளாக. செம்பு மற்றும் தங்க நாணயங்கள் - நாணயங்கள் அச்சடித்தல் - மணி உருவாக்கும் தொழிற்சாலைகள் - கல்மணிகள், கண்ணாடி மணிகள் - சுடுமண் மணிகள் - சங்கு மணிகள் - எலும்புத்துண்டுகள் - தொல்லியல் சான்றுகள் - சிலப்பதிகாரத்தில் மணிகளின் வகைகள்.

அலகு IV வேளாண்மை மற்றும் நீர்ப்பாசனத் தொழில் நுட்பம் 3

அணை, ஏரி, குளங்கள், மதகு - சோழர்காலக் குமுழித் தூம்பின் முக்கியத்துவம் - கால்நடை பராமரிப்பு - கால்நடைகளுக்காக வடிவமைக்கப்பட்ட கிணறுகள் - வேளாண்மை மற்றும் வேளாண்மைச் சார்ந்த செயல்பாடுகள் - கடல்சார் அறிவு - மீன்வளம் - முத்து மற்றும் முத்துக்குளித்தல் - பெருங்கடல் குறித்த பண்டைய அறிவு - அறிவுசார் சமூகம்.

அலகு V அறிவியல் தமிழ் மற்றும் கணித்தமிழ் 3

அறிவியல் தமிழின் வளர்ச்சி - கணித்தமிழ் வளர்ச்சி – தமிழ் நூல்களை மின்பதிப்பு செய்தல் - தமிழ் மென்பொருட்கள் உருவாக்கம் - தமிழ் இணையக் கல்விக்கழகம் - தமிழ் மின் நூலகம் - இணையத்தில் தமிழ் அகராதிகள் - சொற்குவைத் திட்டம்.

TOTAL: 15 PERIODS

TEXT BOOKS CUM REFERENCES

1. தமிழக வரலாறு - மக்களும் பண்பாடும் - கே.கே. பிள்ளை (வெளியீடு:தமிழ்நாடு பாடநூல் மற்றும் கல்வியியல் பணிகள் கழகம்).
2. கணினித் தமிழ்- முனைவர் இல. சுந்தரம். (விகடன் பிரசுரம்).
3. கீழடி - வைகை நதிக்கரையில் சங்ககால நகர நாகரிகம் (தொல்லியல் துறை வெளியீடு)
4. பொருநை - ஆற்றங்கரை நாகரிகம். (தொல்லியல் துறை வெளியீடு)
5. Social Life of Tamils (Dr.K.K.Pillay) A joint publication of TNTB & ESC and RMRL - (in print)
6. Social Life of the Tamils - The Classical Period (Dr.S.Singaravelu) (Published by:International Institute of Tamil Studies.
7. Historical Heritage of the Tamils (Dr.S.V.Subatamanian, Dr.K.D. Thirunavukkarasu)(Published by: International Institute of Tamil Studies).
8. The Contributions of the Tamils to Indian Culture (Dr.M.Valarmathi) (Published by:International Institute of Tamil Studies.)
9. Keeladi - 'Sangam City Civilization on the banks of river Vaigai' (Jointly Published by: Department of Archaeology & Tamil Nadu Text Book and Educational Services Corporation, Tamil Nadu)
10. Studies in the History of India with Special Reference to Tamil Nadu (Dr.K.K.Pillay)(Publishedby: The Author)
11. Porunai Civilization (Jointly Published by: Department of Archaeology & Tamil Nadu Text Book and Educational Services Corporation, Tamil Nadu)
12. Journey of Civilization Indus to Vaigai (R.Balakrishnan) (Published by: RMRL) – Reference Book.

24ES201 **BASIC ELECTRICAL AND ELECTRONICS ENGINEERING**

Offered by EEE (Common to CE, AI&DS, CSE, CSE(AIML), CSBS, IT & MECH branches) (II Sem.)

L	T	P	C
3	0	2	4

COURSE OBJECTIVES

- To introduce the basics of DC circuits and analysis
- To introduce the basics of AC circuits and analysis
- To introduce analog devices and their characteristics
- To educate on the fundamental concepts of digital electronics
- To introduce the functional elements and working of measuring instruments and Electrical Machines

UNIT I DC CIRCUITS **9**

Circuit Components: Conductor, Resistor, Inductor, Capacitor – Ohm's Law - Kirchhoff's Laws – Independent and Dependent Sources - Power and energy - Mesh analysis with independent sources only (Steady state).

UNIT II AC CIRCUITS **9**

Introduction to AC Circuits and Parameters: Waveforms, Average value, RMS Value, Instantaneous power, real power, reactive power and apparent power, power factor – Steady state analysis of RLC circuits (Simple problems only).

UNIT III ANALOG ELECTRONICS **9**

Resistors colour coding - Semiconductor Devices: PN Junction Diodes, Zener Diode –Characteristics Applications – Bipolar Junction Transistor, JFET, SCR, MOSFET - I-V Characteristics and Applications, Diode Rectifiers.

UNIT IV DIGITAL FUNDAMENTALS **9**

Review of number systems, binary codes, error detection and correction codes, Combinational logic - representation of logic functions-SOP and POS forms, K-map representations - minimization using K maps (Simple Problems only).

UNIT V MACHINES AND MEASUREMENTS **9**

Moving Coil meters, Moving Iron meters, Energy Meter, DSO and Data acquisition system. Faradays Laws, Lenz's Law, Fleming's Rules, Statically and dynamically induced EMF; Construction and Working principle of three phase Induction Motor, Single phase Transformer, Stepper Motor, Servo Motor.

EXPERIMENTS (ANY FIVE):

1. Verification of ohm's Law and Kirchhoff's laws
2. V-I characteristics of PN junction diode and Zener diode
3. Full wave Diode Rectifier
4. Verification of Logic gates
5. Load test on Three phase Induction Motor
6. Load Test on single phase Transformer
7. Characteristics of BJT-CE configuration

TOTAL: 45+15 PERIODS

TEXT BOOKS

1. Kothari DP and I.J Nagrath, "Basic Electrical and Electronics Engineering", Second Edition, McGraw Hill Education, 2020.
2. S.K. Bhattacharya "Basic Electrical and Electronics Engineering", Pearson Education, Second Edition, 2017.
3. Sedha R.S., "A text book book of Applied Electronics", S. Chand & Co., 2008.
4. James A. Svoboda, Richard C. Dorf, "Dorf's Introduction to Electric Circuits", Wiley, 2018.
5. A.K. Sawhney, Puneet Sawhney 'A Course in Electrical & Electronic Measurements & Instrumentation', Dhanpat Rai and Co, 2015.

REFERENCES

1. Kothari DP and I.J Nagrath, “Basic Electrical Engineering”, Fourth Edition, McGraw Hill Education, 2019.
2. Thomas L. Floyd, ‘Digital Fundamentals’, 11th Edition, Pearson Education, 2017.
3. Albert Malvino, David Bates, ‘Electronic Principles, McGraw Hill Education; 7th edition, 2017.
4. Mahmood Nahvi and Joseph A. Edminister, “Electric Circuits”, Schaum’ Outline Series, McGraw Hill, 2002.
5. H.S. Kalsi, ‘Electronic Instrumentation’, Tata McGraw-Hill, New Delhi, 2010.

COURSE OUTCOMES

After the completion of the course, the students will be able to

CO1: Compute Electric DC Circuit parameters for simple problems

CO2: Compute the AC parameters for simple problems

CO3: Analyze the characteristics of analog electronic devices

CO4: Explain the basic concepts of digital electronics

CO5: Explain the operating principles of measuring instruments

CO6: Explain the working principle and applications of electrical machines

CO’s-PO’s & PSO’s MAPPING

COs	POs												PSOs		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
1	3	3	3	2	2	1	-	-	2	-	2	2	2	2	3
2	3	3	3	2	2	1	-	-	2	-	2	2	2	2	3
3	3	3	3	2	2	1	-	-	2	-	2	2	2	2	3
4	3	3	3	2	2	1	-	-	2	-	2	2	2	2	3
5	3	3	3	2	2	1	-	-	2	-	2	2	2	2	3
6	3	3	3	2	2	1	-	-	2	-	2	2	2	2	3
Avg.	3	3	3	2	2	1	-	-	2	-	2	2	2	2	3

1 - Low, 2 - medium, 3 - high, ‘-“- no correlation

24CS211 PROGRAMMING IN C LABORATORY

Offered by CSE (Common to AIDS, CSE, CSE(AIML), CSBS & IT branches) (II Sem.)

L	T	P	C
0	0	4	2

COURSE OBJECTIVES

- To familiarize with C programming constructs
- To develop programs in C using basic constructs
- To develop programs in C using arrays
- To develop applications in C using strings, pointers, functions
- To develop applications in C using structures
- To develop applications in C using file processing

EXPERIMENTS:

Note: The examples suggested in each experiment are only indicative. The lab instructor is expected to design based on the topics listed. The Examination shall not be restricted to the sample experiments listed here.

1. Programs using I/O statements and conditional constructs (if, switch, ternary)
2. Programs using looping constructs (for, while, do-while)
3. Programs using functions with different parameter passing techniques: Call by value (e.g. swapping two numbers), call by reference (e.g. changing the elements of an array), Recursion (e.g. binary search)
4. Programs using one dimensional array (e.g. inserting an element after every i^{th} position in an array, Insertion sort)
5. Programs using multi-dimensional arrays (e.g. matrix manipulations)
6. Programs using strings and their operations (e.g. concatenation of strings, extracting a substring, checking for palindrome, search for a given string using binary search)
7. Programs to demonstrate pointers to functions (e.g. simple arithmetic calculator)
8. Programs to demonstrate simple structure manipulations (e.g. generating a transcript with CGPA and class obtained)
9. Programs to pass structures to functions (e.g. operations on complex numbers, difference between times)
10. Programs to pass array of structures to a function (e.g. generate invites to N parents for a meeting by passing the details of students)
11. Programs to demonstrate file operations (e.g. count the number of characters, words and lines in a file, replace a specific word with the given word in the same file)

MINI PROJECTS

Develop an application modularly using C programming constructs. (eg. Library management system, Online ticket reservation system – Train/Bus/Airways, Hotel reservation system)

TOTAL: 60 PERIODS

COURSE OUTCOMES:

Upon completion of the course, the students will be able to

CO1: Demonstrate knowledge on C Programming constructs
CO2: Apply appropriate control structures to solve problems
CO3: Design and implement applications using arrays and strings
CO4: Develop user defined functions and apply concept of recursion to solve problems
CO5: Develop applications in C using structures and pointers
CO6: Implement operations on files

TEXT BOOKS:

1. E. Balaguruswamy, "Programming in ANSI C", 8th Edition, 2019, McGraw Hill Education, ISBN: 978-93-5316-513-0.
2. Yashavant P. Kanetkar, "Let Us C", 19th Edition, 2022, BPB Publications, ISBN 13: 978-9355512765.

REFERENCE BOOKS:

1. Kernighan B.W and Dennis M. Ritchie, "The C Programming Language", 2nd Edition, 2015, Pearson Education India, ISBN: 978-93-3254-944-9.

2. Reema Thareja, "Programming in C", Oxford university press, Second Edition, 2016, ISBN: 9780199456147
3. Anita Goel, Ajay Mittal, “Computer Fundamentals and Programming in C”, Pearson Education India , 2016. ISBN: 9789332579200
4. Jacqueline A Jones and Keith Harrow, “Problem Solving with C”, Pearson Education. ISBN: 978-93-325-3800-9.

CO’s-PO’s & PSO’s MAPPING

COs	**POs**												**PSOs**		
	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**	**PSO1**	**PSO2**	**PSO3**
1	3	3	2	2	1	-	-	-	2	-	1	1	3	2	1
2	3	3	2	2	1	-	-	-	2	-	1	1	3	2	1
3	3	3	2	2	1	-	-	-	2	-	1	1	3	2	1
4	3	3	2	2	1	-	-	-	2	-	1	1	3	2	1
5	3	3	2	2	1	-	-	-	2	-	1	1	3	2	1
6	3	3	2	2	1	-	-	-	2	-	1	1	3	2	1
Avg	**3**	**3**	**2**	**2**	**1**	-	-	-	**2**	-	**1**	**1**	**3**	**2**	**1**

1 - low, 2 - medium, 3 - high, ‘-‘ - no correlation

24EM211 **COMMUNICATION LABORATORY II**

Offered by English (Common to ALL branches) (II Sem.)

L	T	P	C
0	0	2	1

COURSE OBJECTIVES

- Equip students with the English language skills required for the successful undertaking of academic studies with primary emphasis on LSRW skills
- Provide guidance and practice in basic general and classroom conversation and to engage in specific academic speaking activities
- Motivate students to speak and present their ideas confidently and effectively
- Improve general and professional reading and writing skills
- Provide more opportunities to develop their project and proposal writing skills
- Make effective presentations by incorporating technical jargons, IC tools, charts, diagrams etc.

UNIT I LISTENING **6**

Listening and take notes of lectures – listening for general information – specific information – listen for details – Speech sounds-vowels and consonants, transcripts.

UNIT II SPEAKING **6**

Ice breakers – JAM – Greetings - Taking leave - Introducing oneself and others - Small talk - Role play - speaking clearly - intonation patterns - converse on everyday topics - sharing memorable incidents – Debate - Turncoat-Group discussion.

UNIT III READING **6**

Read for details – Use of graphic organizer store view and aid comprehension - Reading comprehension - Understanding and inferring - reading for pleasure (novels / short stories / online blogs).

UNIT IV WRITING **6**

Article writing – Review writing – movies / books / journals / blogs – Project / proposal writing –Statement of Purpose (SOP) – Letter of recommendation.

UNIT V EFFECTIVE COMMUNICATION - PRESENTATION **6**

Effective communication – The seven C' s of effective communication - Strategies for presentations - group / pair presentations - making oral presentations - presentations through an aid (Power Point presentations).

TOTAL: 30 PERIODS

COURSE OUTCOMES

After the completion of the course, the students will be able to

CO1: To improve listening skill

CO2: To participate in group discussions confidently and appropriately and inconversations both formal and informal

CO3: To enhance reading skill and comprehend

CO4: To writing articles improving the art of writing letter of recommendation and SOPs

CO5: To develop more proficient communication abilities

CO6: To make effective presentations

CO’s-PO’s & PSO’s MAPPING

COs	POs												PSOs		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
1	-	-	-	-	-	-	-	2	3	3	-	1	-	-	-
2	-	-	-	-	-	-	-	2	3	3	-	1	-	-	-
3	-	-	-	-	-	-	-	2	3	3	-	1	-	-	-
4	-	-	-	-	-	-	-	2	3	3	-	1	-	-	-
5	-	-	-	-	-	-	-	2	3	3	-	1	-	-	-
6	-	-	-	-	-	-	-	2	3	3	-	1	-	-	-
Avg.	-	-	-	-	-	-	-	2	3	3	-	1	-	-	-

1 - Low, 2 - medium, 3 - high, ‘-“- no correlation

24MA301 PROBABILITY AND STATISTICS
Offered by Maths (Common to AIDS, CSE, CSE(AIML), CSBS & IT) (III Sem.)

L	T	P	C
3	1	0	4

COURSE OBJECTIVES:

- Introduce the fundamental concepts of random variables and moments.
- Gain the knowledge on standard distributions which are used in real time problems.
- Familiarize the basic concepts of two-dimensional random variables and transformation of functions.
- Acquaint students with knowledge on hypothesis testing for small and large samples, which is crucial for real-world phenomena.
- Grasp the preliminary concepts of various non-parametric tests.
- Gain insight into the fundamental concepts and description of statistical quality control, which are extensively applied in real-world problems.

UNIT I RANDOM VARIABLES AND DISTRIBUTIONS 9 + 3

Random variables – Discrete and continuous randomvariables – Moments – Moment generating functions – Binomial, Poisson, Geometric, Uniform, Exponential and Normal distributions.

UNIT II TWO- DIMENSIONAL RANDOM VARIABLES 9 + 3

Joint distributions – Marginal and conditional distributions – Covariance – Correlation and linear regression – Discrete and continuous cases – Transformation of random variables.

UNIT III TESTING OF HYPOTHESIS - PARAMETIRC TESTS 9 + 3

Sampling distributions – Large sample tests for single mean, single proportion and difference of means – Small sample tests for single mean and difference of means – t-test for paired data – F - test for equality of variances.

UNIT IV TESTING OF HYPOTHESIS - NON- PARAMETRIC TESTS 9 + 3

Chi square tests for goodness of fit and independence of attributes – The Sign test – Rank – sum tests – The U test – The H test – Tests based on Runs – Test of randomness – The Kolmogorov – Smirnov test.

UNIT V STATISTICAL QUALITY CONTROL 9 + 3

Control charts for measurements ($\overline{X}$ and R charts) – Control charts for attributes (p, c and np charts) – Tolerance limits – Acceptance sampling.

TOTAL: 60 PERIODS

COURSE OUTCOMES:
After the completion of the course, the students will be able to
CO1: Understand the fundamental concepts of random variables and their moments.
CO2: Acquaint the knowledge on standard distributions that describe real-life phenomena.
CO3: Comprehend the basic concepts of two-dimensional random variables and apply them to engineering applications.
CO4: Utilize hypothesis testing for small and large samples in real-life situations.
CO5: Acquire knowledge of non-parametric tests, widely used in real-world problems.
CO6: Gain insight into statistical quality control methods applicable to challenging issues in solutions.

TEXT BOOKS:

[1] Richard A. Johnson, Irwin Miller, John E Freund, *Miller & Freund's Probability and Statistics for Engineers,* 9th Edition, Pearson Education, 2018.
[2] Irwin Miller, Marylees Miller, *John E. Freund's Mathematical Statistics with Applications,* 8th Edition, Pearson Education, 2014.

[3] Douglas C. Montgomery – *Introduction to Statistical Quality Control,* 8th Edition, Wiley Publications, 2019.

REFERENCES:

[1] Gupta, S.C., Kapoor, V.K., *Fundamentals of Mathematical Statistics*, 12th Edition, Sultan Chand and Sons, New Delhi, 2020.
[2] Jay L. Devore, *Probability and Statistics for Engineering and the Sciences*, 9th Edition, Cengage India Private Limited, 2016.
[3] Sheldon M. Ross, *Introduction to Probability and Statistics for Engineers and Scientists*, 6th Edition, Academic Press, 2021.
[4] Spiegel, M.R., Schiller, J., Srinivasan, R.A., *Schaum's Outlines on Probability and Statistics,* 4th Edition, Tata McGraw Hill Edition, 2013.
[5] Walpole R.E., Myers R.H., Myers S.L., Ye K., *Probability & Statistics for Engineers & Scientists,* Global Edition, Pearson Education, 2016.

Mapping of COs with POs and PSOs:

The mapping between the COs, POs and PSOs is given in the following table.

COs	Programme Outcomes (POs)												PSOs		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	2	2	-	-	-	-	2	-	-	2	-	-	-
CO2	3	3	2	2	-	-	-	-	2	-	-	2	-	-	-
CO3	3	3	2	2	-	-	-	-	2	-	-	2	-	-	-
CO4	3	3	2	2	-	-	-	-	2	-	-	2	-	-	-
CO5	3	3	2	2	-	-	-	-	2	-	-	2	-	-	-
CO6	3	3	2	2	-	-	-	-	2	-	-	2	-	-	-
Avg.	3	3	2	2	-	-	-	-	2	-	-	2	-	-	-

1 - low, 2 - medium, 3 - high, '-"- no correlation

24CS301 DATA STRUCTURES
Offered by CSE (Common to CSE, CSE(AIML), CSBS & IT) (III Sem.)

L	T	P	C
3	0	0	3

COURSE OBJECTIVES:

- To acquire knowledge on ADTs such as List, Stack and Queue
- To apply the linear and non-linear data structures to various problems
- To explore the binary trees and its traversals
- To know the concepts of graph, its representation, traversal and applications
- To learn sorting and hashing techniques

UNIT I LINEAR DATA STRCTURES - LIST 9
Introduction to Data Structures - Abstract Data Types (ADTs) - List ADT: Array implementation of lists – Linked lists – Circular linked lists – Doubly linked lists – Applications of Lists: Polynomial manipulation.

UNIT II LINEAR DATA STRUCTURES – STACKS AND QUEUES 9
Stack ADT: Array Implementation of stacks – Linked List Implementation of stacks - Applications: Balancing symbols – Infix to postfix conversion – Evaluating postfix expressions – Queue ADT: Array implementation of queues – Linked List Implementation of queues - Circular Queue – Double Ended Queue- Applications of queues: FCFS Scheduling.

UNIT III NON-LINEAR DATA STRUCTURES - TREES 9
Tree Terminologies - Binary trees - Expression trees - Binary search tree ADT - AVL trees – Insertion and Deletion Operation - Tree traversals - Priority Queue (Heap).

UNIT IV NON LINEAR DATA STRUCTURES - GRAPHS 9
Graph Definitions – Representation of graphs - Graph Traversals: Breadth-first traversal – Depth-first traversal - Topological sort – Shortest-Path Algorithms: Dijkstra's algorithm –Minimum Spanning Tree: Prim's Algorithm – Kruskal's Algorithm

UNIT V SORTING AND HASING TECHNIQUES 9
Sorting: Selection sort – Bubble sort - Insertion sort – Shell sort – Radix Exchange sort, Heap sort; Hashing: Hash function – Separate chaining – Open addressing – Rehashing – Extendible hashing.

TOTAL: 45 PERIODS

TEXT BOOKS

1. M A Weiss, "Data Structures and Algorithm Analysis in C", 2nd Edition, Pearson Education, 2002.
2. 2. Richard F Gilberg, Behrouz A Forouzan, "Data Structures: A Pseudocode Approach with C", 2nd Edition, Cengage India, 2007.

REFERENCES

1. A V Aho, J E Hopcroft, J D Ullman, "Data Structures and Algorithms", Pearson Education, New Delhi, Fourteenth Impression, 2013.
2. Ellis Horowitz, Sartaj Sahni, Susan Anderson-Freed, "Fundamentals of Data Structures in C", 2nd Edition, University Press, 2008.
3. Jean-Paul Tremblay and Paul G.Sorenson, "An Introduction to Data Structures with Applications", Tata McGraw-Hill, New Delhi, Second Edition, 2004.

COURSE OUTCOMES:
After the completion of the course, the students will be able to

CO1: Implement and apply various types of lists.
CO2: Implement Stack to solve the specific problems.
CO3: Implement Queue to solve the specific problems.
CO4: Construct different types of trees and implement their operations.
CO5: Represent and apply graph data structure
CO6: Apply sorting and hashing algorithms to solve various computing problems

CO's-PO's & PSO's MAPPING

COs	POs												PSOs		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
1	2	2	3	1	1	-	-	-	1	1	-	2	3	2	1
2	2	2	3	1	1	-	-	-	1	1	-	2	3	2	1
3	2	2	3	1	1	-	-	-	1	1	-	2	3	2	1
4	2	2	3	2	1	-	-	-	1	1	-	2	3	2	1
5	2	2	3	2	1	-	-	-	1	1	-	2	3	2	1
6	2	2	3	1	1	-	-	-	1	1	-	2	3	2	1
Avg.	2	2	3	1	1	-	-	-	1	1	-	2	3	2	1

1 - low, 2 - medium, 3 - high, '-"- no correlation

24CS302 **OBJECT ORIENTED PROGRAMMING**
Offered by CSE (Common to AIDS, CSE, CSE(AIML), CSBS & IT) (III Sem.)

L	T	P	C
3	0	0	3

COURSE OBJECTIVES:

- The objective of this course is to provide object oriented concepts through which robust, Secured and reusable software can be developed.
- To understand object oriented principles like abstraction, encapsulation, inheritance, polymorphism and apply them in solving problems.
- To understand the implementation of packages and interfaces.
- To understand the concepts of exception handling, multithreading and collection classes.
- To understand how to connect to the database using JDBC.
- To understand the design of Graphical User Interface using JavaFX.

UNIT I INTRODUCTION 9

Java Programming- comments, Data types, Variables, Constants, Scope and Lifetime of variables, Operators, Type conversion and casting, Enumerated types, Control flow- block scope, conditional statements, loops, break and continue statements, arrays, simple java stand alone programs, class, object, and its methods constructors, methods, static fields and methods, access control, this reference, overloading constructors, recursion, exploring string class, garbage collection.

UNIT II OBJECT ORIENTED PRINCIPLES 9

Inheritance – Inheritance types, super keyword, preventing inheritance: final classes and methods. Polymorphism – method overloading and method overriding, abstract classes and methods. Interfaces- Interfaces Vs Abstract classes, defining an interface, implement interfaces, accessing implementations through interface references, extending interface, inner class. Packages- Defining, creating and accessing a package, importing packages.

UNIT III EXCEPTION HANDLING, MULTITHREADING 9

Exception handling-Benefits of exception handling, the classification of exceptions - exception hierarchy, checked exceptions and unchecked exceptions, usage of try, catch, throw, throws and finally, creating own exception subclasses. Multithreading – Differences between multiple processes and multiple threads, thread life cycle, creating threads, interrupting threads, thread priorities, synchronizing threads, inter-thread communication, producer consumer problem.

UNIT IV COLLECTIONS, STRINGS , I/O STREAMS 9

Collection Framework in Java – Introduction to java collections, Overview of java collection framework, Commonly used collection classes: Array List, Vector, Hash table, Stack. Java string methods – mutable and immutable strings.

I/O Streams- Byte streams, Character streams, Text input/output, Binary input/output, File management using File class.

UNIT V GUI PROGRAMMING, JDBC 9

JavaFx – components -Layout management – Event Handling- Events, Event sources, Event classes, Event Listeners, Delegation event model, Examples: Handling Mouse and Key events. Connecting to Database – JDBC drivers.

TOTAL: 45 PERIODS

TEXT BOOK:

1. Java Fundamentals–A Comprehensive Introduction, Herbert Schildt and DaleSkrien, TMH, 2017.
2. JavaFX A Beginners Guide - J. F. DiMarzio, Mcgraw Hill, 2011

REFERENCES:

1. Java for Programmers, P.J.Deitel and H.M.Deitel, PEA (or) Java: How to Program ,P.J.Deitel and H.M.Deitel,PHI, 2009.

2. Core Java: An Integrated Approach – Dr R Nageswara Rao, Dreamtech press , 2016

3. Object Oriented Programming through Java -P.Radha Krishna,Universities Press, 2006

4. Java: The Complete Reference - Herbert Schildt ,Dr. Coward, Danny, Mcgraw hill, 2024

5. Programming in Java, S. Malhotra and S. Choudhary, Oxford University Press.

COURSE OUTCOMES:

After the completion of the course, the students will be able to
CO1: Learn the basic constructs of Java Programming
CO2: Apply the features of Object Oriented Programming
CO3: Handle exceptions while running application
CO4: Understand the concept of multithreading
CO5: Construct various basic data structures using the collection framework
CO6: Learn how to create GUI applications

CO's-PO's & PSO's MAPPING

COs	POs												PSOs		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
1	3	3	3	-	2	1	-	-	1	1	-	2	2	2	2
2	3	3	3	-	2	1	-	-	1	1	-	2	2	2	2
3	3	3	3	2	2	1	-	-	1	1	-	2	2	2	2
4	3	3	3	2	2	1	-	-	1	1	-	2	2	2	2
5	3	3	3	2	2	1	-	-	1	1	-	2	2	2	2
6	3	3	3	2	2	1	-	-	1	1	-	2	2	2	2
Avg.	**3**	**3**	**3**	**2**	**2**	**1**	**-**	**-**	**1**	**1**	**-**	**2**	**2**	**2**	**2**

1 - low, 2 - medium, 3 - high, '-"- no correlation

24ES301 DIGITAL PRINCIPLES AND COMPUTER ORGANIZATION
Offered by CSE (Common to AIDS, CSE, CSE(AIML) & CSBS) (III Sem.)

L	T	P	C
3	0	2	4

COURSE OBJECTIVES:

- To analyze and design combinational circuits.
- To analyze and design sequential circuits
- To understand the basic structure and operation of a digital computer.
- To study the design of data path unit, control unit for processor and to familiarize with the hazards.
- To understand the concept of various memories and I/O interfacing.

UNIT I COMBINATIONAL LOGIC 9

Combinational Circuits – Karnaugh Map - Analysis and Design Procedures – Binary Adder – Subtractor – Decimal Adder - Magnitude Comparator – Decoder – Encoder – Multiplexers – Demultiplexers

UNIT II SYNCHRONOUS SEQUENTIAL LOGIC 9

Introduction to Sequential Circuits – Flip-Flops – operation and excitation tables, Triggering of FF, Analysis and design of clocked sequential circuits – Design – Moore/Mealy models, state minimization, state assignment, circuit implementation - Registers – Counters.

UNIT III COMPUTER FUNDAMENTALS 9

Functional Units of a Digital Computer: Von Neumann Architecture – Operation and Operands of Computer Hardware Instruction – Instruction Set Architecture (ISA): Memory Location, Address and Operation – Instruction and Instruction Sequencing – Addressing Modes, Encoding of Machine Instruction – Interaction between Assembly and High Level Language.

UNIT IV PROCESSOR 9

Instruction Execution – Building a Data Path – Designing a Control Unit – Hardwired Control, Micro programmed Control – Pipelining – Data Hazard – Control Hazards.

UNIT V MEMORY AND I/O 9

Memory Concepts and Hierarchy – Memory Management – Cache Memories: Mapping and Replacement Techniques – Virtual Memory – DMA – I/O – Accessing I/O: Parallel and Serial Interface – Interrupt I/O – Interconnection Standards: USB, SATA

TOTAL: 45 PERIODS

LIST OF EXPERIMENTS:

1. Verification of Boolean theorems using logic gates.
2. Design and implementation of combinational circuits using gates for arbitrary functions.
3. Implementation of 4-bit binary adder/ subtractor circuits.
4. Implementation of code converters.
5. Implementation of BCD adder, encoder and decoder circuits
6. Implementation of functions using Multiplexers.
7. Implementation of the synchronous counters
8. Implementation of a Universal Shift register.
9. Simulator based study of Computer Architecture

TOTAL: 30 PERIODS

TOTAL: 45 + 30 = 75 PERIODS

TEXTBOOKS

1. M. Morris Mano, Michael D. Ciletti, "Digital Design: With an Introduction to the Verilog HDL, VHDL, and System Verilog", Sixth Edition, Pearson Education, 2018.
2. David A.Patterson, John L.Hennessy, "Computer Organization and Design, The Hardware/Software Interface", Sixth Edition, Morgan Kaufmann/Elsevier, 2020.

REFERENCES

1. Carl Hamacher, Zvonko Vranesic, Safwat Zaky, Naraig Manjikian, “Computer Organization and Embedded Systems”, Sixth Edition, Tata McGraw-Hill, 2012.
2. William Stallings, “Computer Organization and Architecture Designing for Performance”, Tenth Edition, Pearson Education, 2016.
3. M.Morris Mano, “Digital Logic and Computer Design”, Pearson Education, 2016.

COURSE OUTCOMES:

After the completion of the course, the students will be able to

CO1: Design various combinational digital circuits using logic gates

CO2: Design sequential circuits and analyze the design procedures

CO3: State the fundamentals of computer systems and analyze the execution of an instruction

CO4: Analyze different types of control design and identify hazards

CO5: Identify the characteristics of various memory systems

CO6: Analyze the characteristics of I/O communication

CO’s-PO’s & PSO’s MAPPING

COs	POs												PSOs		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
1	3	3	3	2	2	1	-	-	3	1	-	2	1	2	-
2	3	3	3	2	2	1	-	-	3	1	-	2	1	2	-
3	3	3	3	2	2	1	-	-	3	1	-	2	1	2	-
4	3	3	3	2	-	1	-	-	3	1	-	2	1	2	-
5	3	3	3	2	-	1	-	-	3	1	-	2	1	2	-
6	3	3	3	2	-	1	-	-	3	1	-	2	1	2	-
Avg.	3	3	3	2	2	1	-	-	3	1	-	2	1	2	-

1 - low, 2 - medium, 3 - high, ‘-“- no correlation

24CS303 **ESSENTIALS OF WEB DEVELOPMENT**
Offered by CSE (Common to CSE, CSE(AIML)) (III Sem.)

L	T	P	C
3	0	2	4

COURSE OBJECTIVES:

- To understand the functioning of the World Wide Web.
- To design and develop web pages using HTML.
- To learn the various types of Cascading Style Sheets.
- To understand and apply core JavaScript concepts including variables, functions, loops, and events.
- To Implement form validation, error handling using javascript.
- To Understand the HTML5 Canvas API and its role in rendering graphics on the web.

UNIT I WORLD WIDE WEB 9

Introduction to Computers and the Internet, Evolution of the Internet and World Wide Web– Internet, Intranet, Extranet-Web Basics: Hyperlinks, URL/URI, DNS - HTTP Request Message – HTTP Response Message – Web-based applications- Web Clients – Web Servers –Web 2.0

UNIT II HTML 9

HTML Introduction - HTML Editors-HTML Comments-HTML Layout-HTML Elements-HTML Headings-HTML Paragraphs-attributes -HTML Text Formatting-HTML Quotations-HTML Color Styles and HSL-HTML Links Hyperlinks-HTML Images-HTML Tables-HTML Lists-HTML Block and Inline Elements. HTML Forms – New HTML 5 FORM Input Types

UNIT III CSS 9

CSS Introduction -CSS Syntax-CSS Selectors- CSS Comments- CSS3 – Inline, embedded and external style sheets – Rule cascading – Inheritance – Backgrounds – Border Images – Colors – Shadows – Text – Transformations – Transitions – Animations.

UNIT IV JAVASCRIPT 9

JavaScript Basics - Variables & Data types - Operators-Statements -Loops -Performance & Debugging-Object-Function-Array-String-Numbers-Math-Objects Java Script Events: Click-DOM: Methods, Document, Elements

UNIT V HTML-CANVAS 9

Introduction - canvas Coordinate System -Rectangles- Using Paths to Draw Lines - Drawing Arcs and Circles - Shadows - Quadratic Curves-Bezier Curves-Linear Gradients - Radial Gradients-Images - Image Manipulation: Processing the Individual Pixels of a canvas - Patterns - Transformations: scale and translate Methods: Drawing Ellipses - rotate Method: Creating an Animation.

TOTAL: 45 PERIODS

TEXT BOOKS:

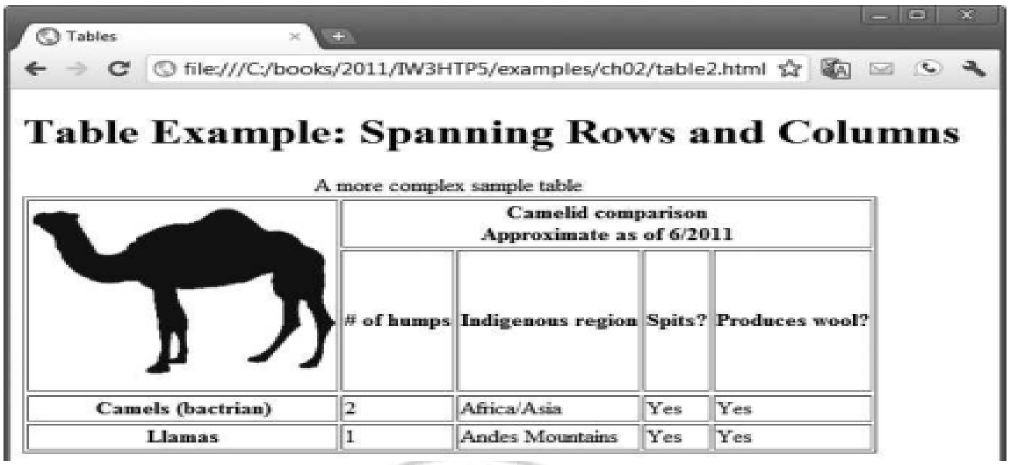
1. Deitel, Harvey Deitel, Abbey Deitel, “Internet and World Wide Web”, 5th edition, Pearson, April 2018.

REFERENCES:

1. Stephen Wynkoop and John Burke, “Running a Perfect Website”, QUE, 2nd Edition, 1999.
2. Chris Bates, Web Programming – Building Intranet Applications, 3rd Edition, Wiley Publications, 2009
3. Jeffrey C and Jackson, Web Technologies A Computer Science Perspective, Pearson Education, 2011.

LIST OF EXPERIMENTS:

1. Create an HTML5 document containing an ordered list of three items—ice cream, soft serve and frozen yogurt. Each ordered list should contain a nested, unordered list of your favourite flavours. Provide three flavours in each unordered list.
2. Create the HTML5 markup that produces a complex table; example is given below.



	Camelid comparison Approximate as of 6/2011			
	# of humps	Indigenous region	Spits?	Produces wool?
Camels (bactrian)	2	Africa/Asia	Yes	Yes
Llamas	1	Andes Mountains	Yes	Yes

3. (Website Registration Form with Optional Survey) Create a website registration form to obtain a user's first name, last name and e-mail address. In addition, include an optional survey question that asks the user's year in college (e.g., Freshman). Place the optional survey question in a details element that the user can expand to see the question.
4. a. (Rounded Corners) Create three div elements, each with a width and height of 100px. On the first element, create slightly rounded corners using a border of 3px black and border-radius of 10px. On the second element, use a border of 3px black and increase the border-radius to 50px.On the third, use a border of 3px black and increase the border-radius to 100px. Make the background-color of each element a different color of your choosing. Inside each element, display the value of the border-radius in bold text.

   b. (Transformation with :hover) Create a transformation program that includes four images. When the user hovers over an image, the size of the image should increase by 20%.
5. a. Write a script that asks the user to enter two numbers, obtains the two numbers from the user and outputs text that displays the sum, product, difference and quotient of the two numbers

   b. Write a script that takes three integers from the user and displays the sum, average, product, smallest and largest of the numbers in an alert dialog.

   c. A large company pays its salespeople on a commission basis. The salespeople receive $200 per week, plus 9 percent of their gross sales for that week. For example, a salesperson who sells $5000 worth of merchandise in a week receives $200 plus 9 percent of $5000, or a total of $650. You have been supplied with a list of the items sold by each salesperson. The values of these items are as follows:

Item	Value
1	239.99
2	129.75
3	99.95

4 350.89

Develop a script that inputs one salesperson's items sold for last week, calculates the salesperson's earnings and outputs HTML5 text that displays the salesperson's earnings.

d. Write function distance that calculates the distance between two points (x1, y1) and (x2,y2). All numbers and return values should be floating-point values. Incorporate this function into a script that enables the user to enter the coordinates of the points through an HTML5 form.

e. (Printing Dates in Various Formats) Dates are printed in several common formats. Write a script that reads a date from an HTML5 form and creates a Date object in which to store it. Then use the various methods of the Date object that convert Dates into strings to display the date in several formats.

f. (Concentric Circles) Write a script that draws eight concentric circles. For each new circle, increase value of the radius by 5. Vary the circles' colors.

g. (Moving Circle) Crate a square canvas with a width and height of 500. Write a script that continuously moves a circle counterclockwise in a diamond pattern so that the circle touches the center of each edge of the canvas.

TOTAL: 30 PERIODS

TOTAL: 45 + 30 = 75 PERIODS

COURSE OUTCOMES:

After the completion of the course, the students will be able to

CO1: Understand functioning of the Internet.
CO2: Design Web Pages using various HTML tags.
CO3: Understand CSS Selectors and apply various types of Cascading Style Sheets.
CO4: Write programs using JavaScript built in objects.
CO5: Understand and apply methods of the Document Object Model
CO6: Draw various Graphical Shapes using HTML Canvas.

CO's-PO's & PSO's MAPPING

COs	POs												PSOs		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
1	2	2	1	1	2	2	-	-	3	2	-	3	1	2	2
2	2	2	1	1	2	2	-	-	3	2	-	3	1	2	2
3	2	2	2	1	2	2	-	-	3	2	-	3	1	2	2
4	2	2	2	1	2	2	-	-	3	2	-	3	1	2	2
5	2	2	2	1	2	2	-	-	3	2	-	3	1	2	2
6	2	2	2	1	2	2	-	-	3	2	-	3	1	2	2
Avg.	**2**	**2**	**2**	**1**	**2**	**2**	**-**	**-**	**3**	**2**	**-**	**3**	**1**	**2**	**2**

1 - low, 2 - medium, 3 - high, '-"- no correlation

24CS304 **FOUNDATIONS OF DATA SCIENCE**

Offered by CSE (Common to CSE, CSE(AIML), CSBS & IT) (III Sem.)

L	T	P	C
3	0	2	4

COURSE OBJECTIVES:

- To understand the data science fundamentals and process.
- To learn to describe the data for the data science process.
- To learn to describe the relationship between data.
- To utilize the Python libraries for Data Wrangling.
- To present and interpret data using visualization libraries in Python.

UNIT I INTRODUCTION 10

Data Science: Benefits and uses - facets of data- Data Science Process: Overview – Defining research goals – Retrieving data – Data preparation - Exploratory Data analysis – build the model – presenting findings and building applications - Basic Statistical descriptions of Data

UNIT II DESCRIBING DATA 8

Types of Data - Types of Variables - Describing Data with Tables and Graphs –Describing Data with Averages - Describing Variability - Normal Distributions.

UNIT III DESCRIBING RELATIONSHIPS 9

Correlation – Scatter plots – correlation coefficient for quantitative data –computational formula for correlation coefficient – Regression –regression line –least squares regression line – Standard error of estimate – multiple regression equations – regression towards the mean

UNIT IV PYTHON LIBRARIES FOR DATA WRANGLING 9

Basics of Numpy arrays – aggregations – computations on arrays – comparisons, masks, boolean logic – fancy indexing – Data manipulation with Pandas – data indexing and selection – operating on data – missing data – combining datasets – aggregation and grouping – pivot tables

UNIT V DATA VISUALIZATION 9

Visualization-Interpreting Data: Importing Matplotlib – Simple Line plots – Simple Scatter plots – visualizing errors – density and contour plots – Exploring Data: Histograms – legends – colors – subplots – text and annotation – customization – three dimensional plotting - Geographic Data with Basemap - Visualization with Seaborn.

TOTAL: 45 PERIODS

LIST OF EXPERIMENTS:

1. Install and explore the features of Jupyter Notebook and packages like numpy, pandas, matplotlib, scipy, statsmodels, sklearn, keras, seaborn, tensorflow.
2. Create and work with data frames using pandas package.
3. Create datasets and explore various commands for doing descriptive analysis.
4. Perform Univariate analysis like frequency, mean, median, mode, variance, standard deviation, skewness and kurtosis using pima Indian diabetes dataset.
5. Perform Bivariate analysis like Linear and logistic regression modeling using datasets.
6. Perform Multiple Regression analysis using pima Indian diabetes dataset.

7. Apply and explore normal curves using pima Indian diabetes dataset.
8. Perform density plotting using predefined datasets.
9. Perform scatter plotting using predefined datasets.
10. Plot histogram, subplots, legend, colors using Iris Dataset.
11. Exploring 3D Plotting Functions using Iris Dataset.
12. Perform visualization of Geographic Data with Basemap.

TOTAL: 30 PERIODS

TOTAL: 45 + 30 = 75 PERIODS

TEXT BOOKS

1. David Cielen, Arno D. B. Meysman, and Mohamed Ali, “Introducing Data Science”, Manning Publications, 2016. (Unit I)
2. Robert S. Witte and John S. Witte, “Statistics”, Eleventh Edition, Wiley Publications, 2017. (Units II and III)
3. Jake VanderPlas, “Python Data Science Handbook”, O’Reilly, 2016. (Units IV and V)

REFERENCES:

1. Allen B. Downey, “Think Stats: Exploratory Data Analysis in Python”, Green Tea Press, 2014.

COURSE OUTCOMES:

After the completion of the course, the students will be able to
CO1: Define the data science process
CO2: Understand different types of data description for data science process
CO3: Gain knowledge on relationships between data
CO4: Use the Python Libraries for Data Wrangling
CO5: Apply visualization Libraries in Python to interpret Data
CO6: Apply visualization Libraries in Python to explore Data

CO’s-PO’s & PSO’s MAPPING

COs	POs												PSOs		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
1	3	3	2	-	2	1	-	-	3	2	-	2	3	2	2
2	3	3	2	-	2	1	-	-	3	2	-	2	3	2	2
3	3	3	2	-	2	1	-	-	3	2	-	2	3	2	2
4	3	3	2	2	2	1	-	-	3	2	-	2	3	2	2
5	3	3	2	2	2	1	-	-	3	2	-	2	3	2	2
6	3	3	2	2	2	1	-	-	3	2	-	2	3	2	2
Avg.	**3**	**3**	**2**	**2**	**2**	**1**	-	-	**3**	**2**	-	**2**	**3**	**2**	**2**

1 - low, 2 - medium, 3 - high, ‘-“- no correlation

24CS311 **DATA STRUCTURES LABORATORY**
Offered by CSE (Common to CSE, CSE(AIML), CSBS & IT) (III Sem.)

L	T	P	C
0	0	3	1.5

COURSE OBJECTIVES:
- To implement different types of List ADT
- Learn how Stack can be implemented using array and Linked list and applications of stack
- Learn how Queue can be implemented using Array and Linked list and applications of queue
- Explore various tree types, traversals, and operations.
- Perform graphs traversals and applications using various algorithms
- Implementing various sorting and hashing techniques

LIST OF EXPERIMENTS
1. To implement Array and Linked List implementations of List ADT
2. To perform polynomial manipulation using List ADT
3. To implement doubly linked list ADT
4. Implementation of Stack
   4a. Array Implementations Stack ADT
   4b. Linked List Implementations Stack ADT
5. To implement various applications of Stack ADT
   5a. Balanced Parenthesis,
   5b. Infix to Postfix Conversion
   5c. Postfix Evaluation
6. Implementation of Queue
   6a. Array Implementations Queue ADT
   6b. Linked List Implementations Queue ADT
7. To implement Double Ended Queue
8. To implement the operations of Binary Search Tree ADT and Tree Traversals
9. To implement the operations of AVL tree
10. To implement Graph Traversals (BFS and DFS)
11. To find out shortest path using Dijkstra's Algorithm
12. To perform sorting (insertion sort, selection sort and bubble sort)
13. To implement Separate chaining hashing techniques

TOTAL: 45 PERIODS

TEXT BOOKS
1. M A Weiss, "Data Structures and Algorithm Analysis in C", 2nd Edition, Pearson Education, 2002.
2. Richard F Gilberg, Behrouz A Forouzan, "Data Structures: A Pseudocode Approach with C", 2nd Edition, Cengage India, 2007.

REFERENCES
1. A V Aho, J E Hopcroft, J D Ullman, "Data Structures and Algorithms", Pearson Education, New Delhi, Fourteenth Impression, 2013.
2. Ellis Horowitz, Sartaj Sahni, Susan Anderson-Freed, "Fundamentals of Data Structures in C", 2nd Edition, University Press, 2008.
3. Jean-Paul Tremblay and Paul G.Sorenson, "An Introduction to Data Structures with Applications", Tata McGraw-Hill, New Delhi, Second Edition, 2004.

COURSE OUTCOMES:
After the completion of the course, the students will be able to

CO1: Implement different types of linked list and its application

CO2: Implement Stack using different data structures and its applications

CO3: Implement types of Queue using different data structures

CO4: Implement Binary Search tree, AVL tree operations

CO5: Implement Graph traversals and shortest path algorithm

CO6: Implement Sorting and Hashing techniques

CO's-PO's & PSO's MAPPING

COs	Pos												PSOs		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
1	2	2	3	1	1	-	-	-	3	2	-	2	3	2	1
2	2	2	3	1	1	-	-	-	3	2	-	2	3	2	1
3	2	2	3	1	1	-	-	-	3	2	-	2	3	2	1
4	2	2	3	2	1	-	-	-	3	2	-	2	3	2	1
5	2	2	3	2	1	-	-	-	3	2	-	2	3	2	1
6	2	2	3	1	1	-	-	-	3	2	-	2	3	2	1
Avg.	2	2	3	1	1	-	-	-	3	2	-	2	3	2	1

1 - low, 2 - medium, 3 - high, '-"- no correlation

24CS312 **OBJECT ORIENTED PROGRAMMING LABORATORY**
Offered by CSE (Common to AIDS, CSE, CSE(AIML), CSBS & IT) (III Sem.)

L	T	P	C
0	0	3	1.5

COURSE OBJECTIVES

- To build software development skills using java programming for real-world applications.
- To understand and apply the concepts of classes, packages, interfaces, inheritance, exception handling and file processing.
- To develop applications using generic programming and event handling

LIST OF EXPERIMENTS

1. Java programs using control structures (sum of even numbers upto100, reversing digits, Pascal triangle, Floyd's Triangle)
2. a. Read the different types of data from the user and create a simple calculator,

   b. Write a Java program to read the different types of data from the user and display that data using command line arguments.
3. Develop stack and queue data structures using classes and objects.
4. Develop a java application with an Employee class with Emp_name, Emp_id, Address, Mail_id, Mobile_no as members. Inherit the classes, Programmer, Assistant Professor, Associate Professor and Professor from employee class. Add Basic Pay (BP) as the member of all the inherited classes with 97% of BP as DA, 10 % of BP as HRA, 12% of BP as PF, 0.1% of BP for staff club funds. Generate pay slips for the employees with their gross and net salary.
5. Write a Java program to create an abstract class BankAccount with abstract methods deposit and withdraw. Create subclasses: SavingsAccount and CurrentAccount that extend the BankAccount class and implement the respective methods to handle deposits and withdrawals for each account type.
6. Write a Java program to create an interface Shape with the getArea(method. Create three classes Rectangle, Circle, and Triangle that implement the Shape interface. Implement the getArea method for each of the three classes.
7. Write a Java program to create a method that takes an integer as a parameter and throws an exception if the number is odd.
8. Write a java program that implements a multi-threaded application that has three threads. First thread generates a random integer every 1 second and if the value is even, the second thread computes the square of the number and prints. If the value is odd, the third thread will print the value of the cube of the number.
9. Write a java program to (i) read from a file (ii) write into a file (iii) append text to a file
10. Develop applications to demonstrate the features of generics classes.
11. Write Java program to create an Arraylist, iterate through it , insert an element in the 'n' position.
12. Develop applications using JavaFX controls, layouts and menus.
13. Develop a mini project for any application using Java concepts

TOTAL: 45 PERIODS

TEXTBOOKS

1. Herbert Schildt,“ Java the complete reference”,9th edition, McGrawHill, Education, 2014.
2. T.Budd,“ Understanding Object-Oriented Programming with Java”, updated edition, Pearson Education, 2000.

REFERENCES

1. J.Ninoand F.A.Hosch,“An Introduction to programming and OO design using Java”, 3rdedition, John Wiley & sons, 2008
2. P.RadhaKrishna,“Object Oriented Programming through Java”,1stedition,Universities Press, 2007.
3. R.A. Johnson, “Java Programming and Object oriented Application Development”, 1st edition, Cengage Learning, 2006.

COURSE OUTCOMES:

After the completion of the course, the students will be able to

CO1: Develop simple applications using java constructs

CO2: Develop java programs using object oriented programming concepts

CO3: Implement multithreading, and generics concepts

CO4: Implement java programs with IO and exception handling

CO5: Implement programs using the collection class

CO6: Create GUIs and event driven programming applications for real world problems

CO’s-PO’s & PSO’s MAPPING

COs	Pos												PSOs		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
1	3	3	3	-	2	1	-	-	3	2	-	2	2	2	2
2	3	3	3	-	2	1	-	-	3	2	-	2	2	2	2
3	3	3	3	2	2	1	-	-	3	2	-	2	2	2	2
4	3	3	3	2	2	1	-	-	3	2	-	2	2	2	2
5	3	3	3	2	2	1	-	-	3	2	-	2	2	2	2
6	3	3	3	2	2	1	-	-	3	2	-	2	2	2	2
Avg.	**3**	**3**	**3**	**2**	**2**	**1**	**-**	**-**	**3**	**2**	**-**	**2**	**2**	**2**	**2**

1 - low, 2 - medium, 3 - high, ‘-“- no correlation

24EM301 **EMPLOYABILITY SKILLS I**
Offered by English & Maths (Common to ALL branches) (III Sem.)

L	T	P	C
0	0	2	1

COURSE OBJECTIVES:

- To make the students to clear with the problems on numbers
- To understand the basic concepts of probability , problems related to time, speed and distance
- To improve thinking capability and the problem solving skills
- To enhance the students' basic grammatical skills
- To enhance the students' communication skills through email and letters
- To enable the students' conversational skills

UNIT I PROBLEMS ON NUMBERS 6

Introduction – Divisibility – Fractions – Arithmetic Progression – Geometric Progression – Square root of Numbers – Rational and Irrational numbers.

UNIT II BASICS IN STATISTICS 6

Arithmetic Mean, Geometric and Harmonic Means – Median – Mode – Discrete and continuous data – Range.

UNIT III TIME, SPEED AND DISTANCE 6

Introduction – Concept of average and relative speed – Time and work – Relationship between the time and completion of a job – amount of work done – efficiency.

UNIT IV ADVANCED GRAMMAR 6

Parts of speech - Articles, Mixed tenses, Subject-verb Agreement, Error Correction, Comprehension – Reading and Listening.

UNIT V COMMUNICATION 6

Communication – Verbal – Non-verbal, Email writing, Formal letter writing, Communicative Functions – Dialogue writing, Presentation skills – Oral – handling questions.

TOTAL: 30 PERIODS

TEXT BOOK: NIL

REFERENCES

[1] R. S. Aggarwal, *Quantitative Aptitude*, Revised and Enlarged edition, S. Chand Publication, 2017.
[2] P. A. Anand, *Wiley's Quantitative Aptitude*, 1st Edition, 2015.
[3] Mohan Rao, *Quantitative Aptitude*, SciTech Publications India Pvt. Ltd., Chennai, 2017.
[4] Meenakshi Raman and Sangeetha Sharma, *Technical Communication: English Skills for Engineers*, Oxford University Press, 4th Edition, 2022.
[5] Dr. P. C. Das, *Applied English Grammar and Composition,* New Central Book Agency, Howrah, 2019.

COURSE OUTCOMES:

After the completion of the course, the students will be able to
CO1: Enhance problems solving skills on numbers
CO2: Approach the real-time problems in the right way
CO3: Enhance the employability skills
CO4: Improve the writing skill
CO5: Improve the communication skill
CO6: Enhance the conversational skill

CO's-PO's & PSO's MAPPING

COs	POs												PSOs		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
1	1	1	1	1	-	-	-	-	1	-	1	1	-	-	-
2	1	1	1	1	-	-	-	-	1	-	1	1	-	-	-
3	1	1	1	1	-	-	-	-	1	-	1	1	-	-	-
4	-	-	-	-	-	-	-	2	3	3	-	1	-	-	-
5	-	-	-	-	-	-	-	2	3	3	-	1	-	-	-
6	-	-	-	-	-	-	-	2	3	3	-	1	-	-	-
Avg.	0.5	0.5	0.5	0.5	-	-	-	1	2	1.5	0.5	1	-	-	-

1 - low, 2 - medium, 3 - high, '-"- no correlation

24MA401 **DISCRETE MATHEMATICS**
Offered by Maths (Common to AIDS, CSE, CSE(AIML), CSBS & IT) (IV Sem.)

L	T	P	C
3	1	0	4

COURSE OBJECTIVES:

- To extend student's logical and mathematical maturity and ability to deal with abstraction.
- To introduce most of the basic terminologies used in computer science courses and ideas to solve practical problems.
- To understand fundamental concepts of set operations and functions algebraically
- To grasp the preliminary concepts of combinatorics and graph theory.
- To work with relations and investigate their properties.
- To comprehend concepts and notions of graphs, paths and circuits.

UNIT I PROPOSITIONAL CALCULUS 9 + 3

Propositions – Logical connectives – Compound propositions – Conditional and Biconditional propositions – Truth tables – Tautologies and Contradictions – Logical equivalences and implications – De-Morgan's Laws – Normal forms – Principal Conjunctive and disjunctive normal forms – Rules of inference – Arguments – Validity of arguments.

UNIT II PREDICATE CALCULUS 9 + 3

Predicates – Statement functions – Variables – free and bound variables – Quantifiers – Universe of discourse – Logical equivalences and implications for quantifier statements – Theory of inference – Rules of universal specifications and generalization – Validity of arguments.

UNIT III SET THEORY AND FUNCTIONS 9 + 3

Set Operations – Properties – Power set – Inclusion and exclusion principle and its applications – Relations – Graph and Matrix of relation – Partial ordering – Equivalence relations – Partitions – Functions – Types of functions – Composition of relations and functions – Inverse functions.

UNIT IV COMBINATORICS 9 + 3

Mathematical induction – Basics of counting – Counting arguments – Pigeonhole principle – Permutations and combinations – Recursion and Recurrence relations – Solving linearrecurrence relations – Generating function method.

UNIT V GRAPH THEORY 9 + 3

Graphs and graph models – Graph terminology and special types of graphs – Graph isomorphism and Matrix representation of graphs – Connected graphs – Euler graphs – Hamiltonian graphs – Hamiltonian paths and circuits.

TOTAL: 60 PERIODS

COURSE OUTCOMES:

At the end of the course, students will be able to

CO1: Grasp knowledge of the concepts required to think logically and mathematically
CO2: Assess the validity of arguments using formal methods of predicate logic.
CO3: Acquire insight to set theory, relations and its applications.
CO4: Learn the notion on functions and relations.
CO5: Perform combinatorial analysis to solve counting problems and analyze algorithms.
CO6: Understand the concept of graphs and its applications.

TEXT BOOKS

[1] Kenneth H. Rosen, *Discrete Mathematics and its Applications*, 8th Edition, Tata McGraw Hill Publications, 2019.

[2] Tremblay, J.P., Manohar, R,, *Discrete Mathematical Structures with Applications to Computer Science*, 35th Reprint, Tata McGraw Hill Pub. Co. Ltd, New Delhi, 2022.

[3] Venkataraman, M.K., Sridharan, N., Chandrasekaran, N., *Discrete Mathematics*, Reprint, The National Publishing Company, Chennai, 2012.

REFERENCES:

[1] Grimaldi, R.P., *Discrete and Combinatorial Mathematics: An Applied Introduction*, 5th Edition, Pearson Education Asia, Delhi, 2013.

[2] Thomas Koshy, *Discrete Mathematics with Applications*, Elsevier, 2009.

[3] Susanna S. Epp. *Discrete Mathematics with Applications*, Metric Version, 5th Edition, Cengage Publications, 2020.

[4] Gilbert Strang, *Introduction to Linear Algebra*, 6th Edition, Wellesley – Cambridge Press, 2023.

[5] Lipschutz, S., Mark Lipson, *Discrete Mathematics, Schaum's Outlines*, 4th Edition, Tata McGraw Hill Pub. Co. Ltd., New Delhi, 2022.

Mapping of COs with POs and PSOs:

COs	POs												PSOs		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	2	2	-	-	-	-	2	-	-	2	-	-	-
CO2	3	3	2	2	-	-	-	-	2	-	-	2	-	-	-
CO3	3	3	2	2	-	-	-	-	2	-	-	2	-	-	-
CO4	3	3	2	2	-	-	-	-	2	-	-	2	-	-	-
CO5	3	3	2	2	-	-	-	-	2	-	-	2	-	-	-
CO6	3	3	2	2	-	-	-	-	2	-	-	2	-	-	-
Avg.	3	3	2	2	-	-	-	-	2	-	-	2	-	-	-

1 - low, 2 - medium, 3 - high, '-"- no correlation

24CH401 ENVIRONMENTAL SCIENCES AND SUSTAINABILITY

Offered by Chem. (Common to ALL branches) (IV Sem.)

L	T	P	C
2	0	0	2

COURSE OBJECTIVES:

The course will enable learners

- To gain knowledge of the environment and various natural resources.
- To understand the structure and functions of ecosystem and the concept of ecological succession.
- To study the importance of biodiversity by assessing its impact on the human world, its functions and values.
- To identify the scientific and technological solutions to pollution issues and waste management.
- To understand the impacts of various social issues related to environment.
- To comprehend the influence of human population on environment.

UNIT I ENVIRONMENT AND NATURAL RESOURCES 7

Definition, scope and importance of environment, need for public awareness. Natural resources: types - Forest resources: use and over-exploitation, deforestation and its impacts - Water resources: use and over-utilization of water, benefits and problems of dams - Mineral resources: use and exploitation, environmental impacts - Food resources: effects of modern agriculture - Energy sources: renewable and non-renewable resources, use of alternate energy sources – Land resources: land degradation, soil erosion and desertification - Role of an individual in conservation of natural resources.

UNIT II ECOSYSTEM AND BIODIVERSITY 7

Ecosystem: concept, structure and function of an ecosystem, energy flow, food chain, food web and ecological pyramid - Ecological succession. Biodiversity: types, values of biodiversity -India as a mega-diversity nation – Hotspots of biodiversity – Threats to biodiversity – Endangered and endemic species of India – Conservation of biodiversity: In-situ and ex-situ method.

UNIT III POLLUTION AND WASTE MANAGEMENT 7

Pollution: Definition, causes, effects and control measures of (a) Air pollution (b) Water pollution (c) Soil pollution (d) Noise pollution (e) Thermal pollution (f) Nuclear pollution – Case studies - Role of an individual in prevention of pollution. Waste management: municipal solid wastes, e- wastes, and hazardous wastes – Recycling of EV batteries - Occupational Health and Safety Management System (OHASMS), ISO 45001(International standard for occupational health and safety management systems).

UNIT IV SOCIAL ISSUES AND ENVIRONMENT 7

From unsustainability to sustainability – Sustainability Development Goals, UN SDGs-17 SDGs- - Water conservation: rain water harvesting, watershed management - Resettlement and rehabilitation: problems and concerns, case study - Climate change, global warming, acid rain, ozone layer depletion, nuclear accident and holocaust – Disaster management: earth quake, cyclone, flood and landslides Environmental protection act - Public awareness.

UNIT V HUMAN POPULATION AND ENVIRONMENT 7

Population growth, variation among nations, population explosion – Family welfare programme – Human rights - Environment and human health – HIV/AIDS – Value education - Women and child welfare - Role of information technology in environment and human health – Case studies.

TOTAL: 35 PERIODS

TEXTBOOKS

1. Anubha Kaushik and C.P. Kaushik, Perspectives in Environmental Studies, New Age International Publishers, 2nd edition, 2021.
2. Benny Joseph, Environmental Science and Engineering, Tata McGraw-Hill, New Delhi, 2017.
3. Gilbert M. Masters, Introduction to Environmental Engineering and Science, 3rd edition, Pearson Education, 2014.

REFERENCES

1. Dharmendra S. Sengar, Enviromental Law, Prentice Hall of India Pvt. Ltd., New Delhi, 2014.
2. Erach Bharucha, Textbook of Environmental Studies, University Press Pvt. Ltd., Hydrabad, 2025.
3. Tyler Miller G. and Scott E. Spoolman, Environmental Science, Cengage Learing India Pvt. Ltd., New Delhi, 2024.
4. Rajagopalan R, Environmental Studies - From Crisis to Cure, Oxford University Press, 2017.

COURSE OUTCOMES:

After the completion of the course, the students will be able to

CO1: Infer the importance of environment and discuss different natural resources, optimum usage and its significance.
CO2: Explain the concept and functions of ecosystem.
CO3: Recall the various functions, values, threats and conservation of biodiversity.
CO4: Explain different types of pollution and suitable methods to prevent it.
CO5: List the various social issues, environmental protection acts and possible solutions.
CO6: Describe the effects and control of population and role of IT in environment and human health.

CO's-PO's & PSO's MAPPING

Course	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	1	2	1	-	2	3	2	1	2	-	2	-	-	-
CO2	2	1	2	1	-	2	3	2	1	2	-	2	-	-	-
CO3	2	1	2	1	-	2	3	2	1	2	-	2	-	-	-
CO4	2	1	2	1	-	2	3	2	1	2	-	2	-	-	-
CO5	2	1	2	1	-	2	3	2	1	2	-	2	-	-	-
CO6	2	1	2	1	-	2	3	2	1	2	-	2	-	-	-
Avg.	**2**	**1**	**2**	**1**	**-**	**2**	**3**	**2**	**1**	**2**	**-**	**2**	**-**	**-**	**-**

1 - low, 2 - medium, 3 - high, '-"- no correlation

24CS401 THEORY OF COMPUTATION
Offered by CSE (IV Sem.)

L	T	P	C
3	0	0	3

COURSE OBJECTIVES:

- To understand foundations of computation including automata theory
- To construct models of regular expressions and languages.
- To design context free grammar and push down automata
- To understand Turing machines and their capability
- To understand Undecidability and NP class problems

UNIT I AUTOMATA AND REGULAR EXPRESSIONS 9

Need for automata theory - Finite Automata (FA) – Deterministic Finite Automata (DFA) – Non-deterministic Finite Automata (NFA) – Equivalence between NFA and DFA– Finite Automata with Epsilon transitions –Equivalence of NFAs with and without ε-moves - Minimization of DFAs.

UNIT II REGULAR EXPRESSIONS AND LANGUAGES 9

Regular expression – Regular Languages- Equivalence of Finite Automata and regular expressions: Sate elimination method, R_{ij} - Proving languages to be not regular (Pumping Lemma) – Closure properties of regular languages.

UNIT III CONTEXT FREE GRAMMAR AND PUSH DOWN AUTOMATA 9

Hierarchy of languages: Chomsky's - Context-Free Grammar (CFG) and Languages – Derivations and Parse trees – Ambiguity in grammars and languages – Push Down Automata (PDA): Definition – Moves - Instantaneous descriptions -Languages of pushdown automata – Equivalence of pushdown automata and CFG: CFG to PDA-PDA to CFG – Deterministic Pushdown Automata.

UNIT IV NORMAL FORMS AND TURING MACHINES 9

Normal forms for CFG – Simplification of CFG- Chomsky Normal Form (CNF) and Greibach Normal Form (GNF) – Pumping lemma for CFL – Closure properties of Context Free Languages –Turing Machine: Basic model – definition and representation – Instantaneous Description – Language acceptance by TM – TM as Computer of Integer functions.

UNIT V UNDECIDABILITY 9

Unsolvable Problems and Computable Functions –PCP-MPCP- Reduction of TM to PCP- Recursive and recursively enumerable languages – Properties - Machine codes for Turing machine - Universal Turing machine - Tractable and Intractable problems - P and NP complete: Kruskal's algorithm, Travelling Salesman Problem.

TOTAL: 45 PERIODS

TEXT BOOKS

1. Hopcroft J.E., Motwani R. & Ullman J.D., "Introduction to Automata Theory, Languages and Computations", 3rd Edition, Pearson Education, 2008.
2. John C Martin , "Introduction to Languages and the Theory of Computation", 4th Edition, Tata McGraw Hill, 2011.

REFERENCES

1. Harry R Lewis and Christos H Papadimitriou , "Elements of the Theory of Computation", 2nd Edition, Prentice Hall of India, 2015.
2. Peter Linz, "An Introduction to Formal Language and Automata", 6th Edition, Jones & Bartlett, 2016.
3. K.L.P.Mishra and N.Chandrasekaran, "Theory of Computer Science: Automata Languages and Computation", 3rd Edition, Prentice Hall of India, 2006.

COURSE OUTCOMES:

After the completion of the course, the students will be able to

CO1: Construct automata theory using Finite Automata
CO2: Write regular expressions for any pattern
CO3: Design context free grammar and Pushdown Automata
CO4: Construct CFL and Normal Forms
CO5: Design Turing machine for computational functions
CO6: Understand decidable and undecidable problems

CO's-PO's & PSO's MAPPING

COs	POs												PSOs		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
1	3	3	2	1	-	-	-	-	1	-	-	2	3	1	1
2	3	3	2	1	-	-	-	-	1	-	-	2	3	1	1
3	3	3	2	2	-	-	-	-	1	-	-	2	3	1	1
4	3	3	2	2	-	-	-	-	1	-	-	2	3	1	1
5	3	3	2	2	-	-	-	-	1	-	-	2	3	1	1
6	3	3	2	2	-	-	-	-	1	-	-	2	3	1	1
Avg.	3	3	2	2	-	-	-	-	1	-	-	2	3	1	1

1 - low, 2 - medium, 3 - high, '-"- no correlation

24CS402 **DATABASE MANAGEMENT SYSTEMS**
Offered by CSE (Common to CSE, CSE(AIML), CSBS & IT) (IV Sem.)

L	T	P	C
3	0	0	3

COURSE OBJECTIVES:

- To understand the concepts of database
- To represent a database system using ER diagrams
- To design and normalize databases for real time applications
- To devise queries using Relational Algebra, Relational Calculus and SQL
- To evaluate and optimize queries
- To implement queries using various RDMS database packages

UNIT I DATABASE SYSTEM CONCEPTS 9

Database System Concepts and Architecture: Data models, Schemas and instances; Three-Schema architecture and data Independence; Database languages and interfaces; The database system environment; Centralized and Client - Server architectures for DBMS. Conceptual Data modeling and database design: Entity types, Entity sets, Attributes and keys; Relationship types, Relationship sets, Roles and structural constraints; Weak entity types; Relationship types of degree higher than two.

UNIT II RELATIONAL DATABASE DESIGN 9

Relational database design using ER-to-Relational mapping; Mapping EER model constructs to relations. Relational model concepts; Relational model constraints and Relational database schemas. Basic SQL: SQL data definition and data types; Specifying constraints in SQL, Basic retrieval queries in SQL; insert, delete, and update statements in SQL. More complex SQL retrieval queries; Specifying constraints as assertions and actions as triggers; Views (virtual tables) in SQL- DCL, TCL. Relational Algebra: Unary relational operations - select and project; Relational algebra operations from set theory; Binary relational operations - join and division

UNIT III NORMALIZATION FOR RELATIONAL DATABASES 9

Informal design guidelines for relation schemas; Functional dependencies-inference rules, equivalence and minimal cover; Normal forms based on primary keys; 1NF, 2NF, 3NF, Boyce-Codd normal form - multivalued dependency; Join dependencies; Properties of relational decompositions. Query Processing and Optimization: Phases of query processing; Translating SQL Queries into Relational Algebra and other operators; Query trees and Heuristics for query optimization.

UNIT IV INTRODUCTION TO TRANSACTION PROCESSING 9

Introduction to transaction processing; Transaction and system concepts; Desirable properties of transactions; Characterizing schedules based on serializability. Concurrency Control Techniques: Two-phase locking techniques for concurrency control; Concurrency control based on timestamp ordering.

UNIT V INDEXING STRUCTURES AND PHYSICAL DATABASE DESIGN 9

Single level and multi level indexing; Dynamic multi level indexing using B trees and B+ trees. Overview of Physical Storage Media - Storage Interfaces - RAID. Parallel and Distributed Storage: Data Partitioning Dealing with Skew in Partitioning -Replication-Parallel Indexing- Distributed File Systems. Introduction to NoSQL systems: Document-based NoSQL systems and MongoDB.

TOTAL: 45 PERIODS

TEXT BOOKS:

1. Ramez, Elmasri and Shamkant B. Navathe, "Fundamentals of Database Systems", 7th edition, Pearson Education, 2016.

REFERENCES:

1. Raghu Rama Krishnan and Johannes Gehrke, "Database Management Systems", 3rd edition, Tata McGraw Hill, 2013.
2. Abraham Silberschatz, Henry F.Korth and S.Sudarshan, "Database System Concepts", 6th edition, Tata McGraw Hill, 2010.

COURSE OUTCOMES:

After the completion of the course, the students will be able to

CO1: Understand various types of data models and Design database using ER model
CO2: Construct Queries using relational algebra and SQL
CO3: Normalize databases for real time applications
CO4: Understand transaction processing and concurrency control
CO5: Understand various indexing strategies, maintenance of database
CO6: Construct queries using NoSQL databases

CO's-PO's & PSO's MAPPING

COs	POs												PSOs		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
1	2	3	3	-	2	-	-	-	1	-	-	2	3	1	2
2	2	3	3	-	2	-	-	-	1	-	-	2	3	1	2
3	2	3	3	2	2	-	-	-	1	-	-	2	3	1	2
4	2	3	3	2	-	-	-	-	1	-	-	2	3	1	2
5	2	3	3	2	-	-	-	-	1	-	-	2	3	1	2
6	2	3	3	2	-	-	-	-	1	-	-	2	3	1	2
Avg.	2	3	3	2	2	-	-	-	1	-	-	2	3	1	2

1 - low, 2 - medium, 3 - high, '-"- no correlation

24CS403 OPERATING SYSTEMS
Offered by CSE (Common to CSE, CSE(AIML), CSBS & IT) (IV Sem.)

L	T	P	C
3	0	0	3

COURSE OBJECTIVES:

- To understand the basics and functions of operating systems.
- To understand processes and threads
- To analyze scheduling algorithms and process synchronization.
- To understand the concept of deadlocks.
- To analyze various memory management schemes.
- To be familiar with I/O management and file systems.
- To be familiar with the basics of virtual machines and Mobile OS like IOS and Android.

UNIT I OPERATING SYSTEM STRUCTURE AND SERVICES 6

Operating System Overview - Objectives and Functions - Evolution of Operating System; Operating System Structures – Operating System Services - User Operating System Interface - System Calls – System Programs - Design and Implementation

UNIT II PROCESS SCHEDULING 12

Processes - Process Concept - Process Scheduling - Operations on Processes - Inter- process Communication. CPU Scheduling: Preemptive and Non-Preemptive: FCFS, SJF, LJF, HRRN, Priority, SRJF, LRJF, RR, Multiple Queue, Multi-level - Scheduling criteria - Threads - Multithread Models – Threading issues.

UNIT III PROCESS SYNCHRONIZATION AND DEADLOCK 9

Process Synchronization - The Critical-Section problem - Synchronization hardware – Semaphores – Mutex - Classical problems of synchronization - Monitors; Deadlock - Methods for handling deadlocks, Deadlock prevention, Deadlock avoidance, Deadlock detection, Recovery from deadlock.

UNIT IV MEMORY MANAGEMENT 9

Main Memory - Swapping - Contiguous Memory Allocation – Paging - Structure of the Page Table - Segmentation, Segmentation with paging; Virtual Memory -Demand Paging – Copy on Write - Page Replacement - Allocation of Frames –Thrashing.

UNIT V STORAGE MANAGEMENT 9

Mass Storage Structure – Disk Structure - Disk Scheduling and Management; File- System Interface: File concept - Access methods - Directory Structure - Directory organization - File system mounting - File Sharing and Protection- File System Structure –Allocation Methods – Free Space Management- Overview of I/O Systems.

TOTAL: 45 PERIODS

TEXTBOOKS:

1. Abraham Silberschatz, Peter Baer Galvin and Greg Gagne, "Operating System Concepts", 10th Edition, John Wiley and Sons Inc., 2018.
2. Andrew S Tanenbaum,"Modern Operating Systems", Pearson, 5thEdition, 2022 New Delhi.

REFERENCES:

1. Ramaz Elmasri, A.Gil Carrick, David Levine,"Operating Systems –ASpiral Approach", Tata McGraw Hill Edition, 2010.
2. William Stallings,"Operating Systems: Internals and Design Principles",7th Edition, Prentice Hall, 2018.
Achyut S.Godbole, Atul Kahate,"Operating Systems",McGraw Hill Education, 2016.

COURSE OUTCOMES:

After the completion of the course, the students will be able to

CO1: Learn the basic concepts of operating systems

CO2: Analyze various scheduling algorithms.

CO3: Understand process synchronization.

CO4: Learn deadlock detection, prevention and avoidance techniques

CO5: Compare and contrast various memory management schemes.

CO6: Understand the functionality of directory, file systems, I/O systems.

CO's-PO's & PSO's MAPPING

COs	Pos												PSOs		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
1	3	2	2	-	-	-	-	-	1	-	-	2	3	3	1
2	3	3	2	3	2	-	-	-	1	-	-	2	3	3	1
3	3	3	2	3	2	-	-	-	1	-	-	2	3	3	1
4	3	3	2	3	2	-	-	-	1	-	-	2	3	3	1
5	3	3	2	3	2	-	-	-	1	-	-	2	3	3	1
6	3	2	2	2	2	-	-	-	1	-	-	2	3	3	1
Avg.	3	3	2	3	2	-	-	-	1	-	-	2	3	3	1

1 - low, 2 - medium, 3 - high, '-"- no correlation

24CS404 DESIGN AND ANALYSIS OF ALGORITHMS

Offered by CSE (Common to CSE, CSE(AIML), CSBS & IT) (IV Sem.)

L	T	P	C
3	0	2	4

COURSE OBJECTIVES

- To critically analyze the efficiency of alternative algorithmic solutions for the same problem
- To illustrate brute force and divide and conquer design techniques.
- To explain dynamic programming and greedy techniques for solving various problems.
- To apply iterative improvement technique to solve optimization problems
- To examine the limitations of algorithmic power and handling it in different problems.

UNIT I FUNDAMENTALS ALGORITHM 8

Notion of an Algorithm –Fundamentals of the Analysis of Algorithm Efficiency – Analysis Framework - Asymptotic Notations and their properties - Mathematical analysis of Recursive and Non-recursive algorithms –Brute Force: Bubble Sort, Sequential Search. Interpolation search, Pattern search: The naïve string-matching algorithm - Rabin-Karp algorithm - Knuth-Morris-Pratt algorithm

UNIT II EXHAUSTIVE SEARCH AND DIVIDE AND CONQUER 10

Exhaustive Search - Traveling Salesman Problem - Knapsack Problem - Assignment problem. Divide and Conquer Methodology – Merge Sort, Quick Sort, Min Max Algorithm, Multiplication of Large Integers and Strassen's Matrix Multiplication.

UNIT III DYNAMIC PROGRAMMING 10

Dynamic Programming : Principle of optimality - Warshall's and Floyd's algorithms – Optimal Binary Search Trees - Multi stage graph – Matrix Chain Multiplication, Knapsack Problem and Memory functions.

UNIT IV GREEDY TECHNIQUE AND ITERATIVE IMPROVEMENT 8

Greedy Technique –prim's and kruskal's algorithm- Dijkstra's algorithm - Huffman Trees and Codes. Iterative Improvement: The Maximum-Flow Problem – Maximum Matching in Bipartite Graphs- The Stable marriage problem.

UNIT V LIMITATIONS OF ALGORITHM POWER 9

Lower - Bound Arguments - P, NP, NP- Complete and NP Hard Problems problem reduction: TSP – 3-CNF problem. Backtracking – N-Queen problem - Hamiltonian Circuit Problem – Subset Sum Problem. Branch and Bound –- Assignment problem – Knapsack Problem – Traveling Salesman Problem. Approximation Algorithms for NP-Hard Problems – Traveling Salesman Problem – Knapsack problem.

TOTAL: 45 PERIODS

LIST OF EXPERIMENTS:

1. Implement Linear Search. Determine the time required to search for an element. Repeat the experiment for different values of n, the number of elements in the list to be searched and plot a graph of the time taken versus n.

2. Implement recursive Binary Search. Determine the time required to search an element. Repeat the experiment for different values of n, the number of elements in the list to be searched and plot a graph of the time taken versus n.
3. Given a text txt [0...n-1] and a pattern pat [0...m-1], write a function search (char pat [ ], char txt [ ]) that prints all occurrences of pat [ ] in txt [ ]. You may assume that n > m.
4. Implement Merge sort method to sort an array of elements and determine the time required to sort. Repeat the experiment for different values of n, the number of elements in the list to be sorted and plot a graph of the time taken versus n.
5. Implement Quick sort method to sort an array of elements and determine the time required to sort. Repeat the experiment for different values of n, the number of elements in the list to be sorted and plot a graph of the time taken versus n.
6. Implement Floyd's algorithm for the All-Pairs- Shortest-Paths problem.
7. Compute the transitive closure of a given directed graph using Warshall's algorithm. Algorithm Design Techniques
8. Implement Matrix Chain Multiplication.
9. Find the minimum cost spanning tree of a given undirected graph using Prim's algorithm.
10. From a given vertex in a weighted connected graph, develop a program to find the shortest paths to other vertices using Dijkstra's algorithm.
11. Implement N-Queens problem using Backtracking.

The programs can be implemented in C/C++/JAVA/ Python.

TOTAL: 30 PERIODS

TOTAL: 45 + 30 = 75 PERIODS

TEXT BOOKS:

1. Anany Levitin, "Introduction to the Design and Analysis of Algorithms", Third Edition, Pearson Education, 2017.
2. Ellis Horowitz, Sartaj Sahni and Sanguthevar Rajasekaran, Computer Algorithms/ C++, Second Edition, Universities Press, 2007.

REFERENCES:

1. Thomas H.Cormen, Charles E.Leiserson, Ronald L. Rivest and Clifford Stein, "Introduction to Algorithms", Third Edition, PHI Learning Private Limited, 2012.
2. Alfred V. Aho, John E. Hopcroft and Jeffrey D. Ullman, "Data Structures and Algorithms", Pearson Education, Reprint 2006.
3. Harsh Bhasin, "Algorithms Design and Analysis", Oxford university press, 2016.
4. S. Sridhar, "Design and Analysis of Algorithms", Oxford university press, 2014.

COURSE OUTCOMES:

After the completion of the course, the students will be able to

CO1: Analyze the efficiency of algorithms and apply Brute force technique.

CO2: Understand divide and conquer technique.

CO3: Use dynamic programming technique to solve problems.

CO4: Apply greedy technique and iterative improvement technique for optimization.

CO5: Understand lower bound arguments and reduction problems.

CO6: Solve problems using backtracking and branch and bound techniques.

CO's-PO's & PSO's MAPPING

COs	POs												PSOs		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
1	3	3	3	2	1	-	-	-	3	2	-	2	3	2	1
2	3	3	3	2	1	-	-	-	3	2	-	2	3	2	1
3	3	3	3	2	1	-	-	-	3	2	-	2	3	2	1
4	3	3	3	3	1	-	-	-	3	2	-	2	3	2	1
5	3	3	3	3	1	-	-	-	3	2	-	2	3	2	1
6	3	3	3	2	1	-	-	-	3	2	-	2	3	2	1
Avg.	3	3	3	2	1	-	-	-	3	2	-	2	3	2	1

1 - low, 2 - medium, 3 - high, '-"- no correlation

24CS411 **DATABASE MANAGEMENT SYSTEMS LABORATORY**
Offered by CSE (Common to CSE, CSE(AIML), CSBS & IT) (IV Sem.)

L	T	P	C
0	0	3	1.5

COURSE OBJECTIVES:

The objective of this course is to enable the students

- To understand data definitions and data manipulation commands.
- To learn the use of nested and join queries.
- To understand functions, procedures and procedural extensions of data bases.
- To be familiar with the use of a front end tool.
- To understand design and implementation of typical database applications.

LIST OF EXPERIMENTS:

1. Database Design using ER modeling, normalization and Implementation for any application
2. Data Definition Commands, Data Manipulation Commands for inserting, deleting, updating and retrieving Tables
3. Integrity Constraints
4. SQL Functions – Single Row Functions (Numeric, String, Date) and Aggregate Functions Group by clause, Having Clause.
5. Database Querying – Simple queries, Nested queries, Sub queries and Joins
6. Views, Sequences, Synonyms
7. Data Control and Transaction Control statements
8. Database Programming: Implicit and Explicit Cursors
9. Procedures and Functions
10. Triggers
11. Database Connectivity with Front End Tools
12. Simple Queries in MongoDB

TOTAL: 45 PERIODS

TEXT BOOKS:

1. Ramez, Elmasri and Shamkant B. Navathe, "Fundamentals of Database Systems", 7th edition, Pearson Education, 2016.

REFERENCES:

1. Raghu Rama Krishnan and Johannes Gehrke, "Database Management Systems", 3rd edition, Tata McGraw Hill, 2013.
2. Abraham Silberschatz, Henry F.Korth and S.Sudarshan, "Database System Concepts", 6th edition, Tata McGraw Hill, 2010.

COURSE OUTCOMES:
After the completion of the course, the students will be able to
CO1: Understand various types of data models and Design database using ER model
CO2: Construct Queries using relational algebra and SQL
CO3: Create Triggers, Views and Indexes.
CO4: Understand various indexing strategies, maintenance of database
CO5: Construct queries using NoSQL databases
CO6: Design a Front End Application and connect it with a Database.

CO's-PO's & PSO's MAPPING

COs	POs												PSOs		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
1	2	3	3	-	2	-	-	-	3	2	-	2	3	1	2
2	2	3	3	-	2	-	-	-	3	2	-	2	3	1	2
3	2	3	3	2	2	-	-	-	3	2	-	2	3	1	2
4	2	3	3	2	2	-	-	-	3	2	-	2	3	1	2
5	2	3	3	2	2	-	-	-	3	2	-	2	3	1	2
6	2	3	3	2	2	-	-	-	3	2	-	2	3	1	2
Avg.	2	3	3	2	2	-	-	-	3	2	-	2	3	1	2

1 - low, 2 - medium, 3 - high, '-"- no correlation

24CS412 OPERATING SYSTEMS LABORATORY
Offered by CSE (Common to CSE, CSE(AIML), CSBS & IT) (IV Sem.)

L	T	P	C
0	0	3	1.5

COURSE OBJECTIVES:
The objective of this course is to enable the students

- To install windows operating systems.
- To understand the basics of Unix command and shell programming.
- To implement various CPU scheduling algorithms.
- To implement Deadlock Avoidance and Deadlock Detection Algorithms
- To implement Page Replacement Algorithms
- To implement various memory allocation methods.
- To be familiar with File Organization and File Allocation Strategies.

LIST OF EXPERIMENTS:

1. Illustrate UNIX commands and Shell Programming (Mean and Std. deviation, Sum of digits, Reversing a number, Command line arguments)
2. Process Management using System Calls: Fork, Exit, Getpid, Wait, Close
3. Write C programs to implement the various CPU Scheduling Algorithms
   a. FCFS b. SJF c. Priority d. RR
4. Inter process communication: Message Passing
5. Implement mutual exclusion by Semaphore
6. Write C programs to avoid Deadlock using Banker's Algorithm
7. Write a C program to Implement Deadlock Detection Algorithm
8. Write C programs to implement the following Memory Allocation Methods
   a. First Fit b. Worst Fit c. Best Fit
9. Write C programs to implement the various Page Replacement Algorithms
10. Write C programs to demonstrate the various File Organization Techniques
11. Implement the following File Allocation Strategies using C programs
    a. Sequential b. Indexed c. Linked

TOTAL: 45 PERIODS

TEXTBOOKS:

1. Abraham Silberschatz, Peter Baer Galvin and Greg Gagne, "Operating System Concepts", 10th Edition, John Wiley and Sons Inc., 2018.
2. Andrew S Tanenbaum,"Modern Operating Systems", Pearson, 5thEdition, 2022 New Delhi.

REFERENCES:

1. Ramaz Elmasri, A.Gil Carrick, David Levine,"Operating Systems –ASpiral Approach", Tata McGraw Hill Edition, 2010.
2. William Stallings,"Operating Systems: Internals and Design Principles",7th Edition, Prentice Hall, 2018.
   Achyut S.Godbole, Atul Kahate,"Operating Systems",McGraw Hill Education, 2016.

COURSE OUTCOMES:

After the completion of the course, the students will be able to

CO1: Learn the basic concepts of operating systems

CO2: Analyze various scheduling algorithms.

CO3: Understand process synchronization.

CO4: Learn deadlock detection, prevention and avoidance techniques

CO5: Compare and contrast various memory management schemes.

CO6: Understand the functionality of directory, file systems, I/O systems.

CO's-PO's & PSO's MAPPING

COs	POs												PSOs		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
1	3	2	2	-	-	-	-	-	3	2	-	2	3	3	1
2	3	3	2	3	2	-	-	-	3	2	-	2	3	3	1
3	3	3	2	3	2	-	-	-	3	2	-	2	3	3	1
4	3	3	2	3	2	-	-	-	3	2	-	2	3	3	1
5	3	3	2	3	2	-	-	-	3	2	-	2	3	3	1
6	3	2	2	2	2	-	-	-	3	2	-	2	3	3	1
Avg.	3	3	2	3	2	-	-	-	3	2	-	2	3	3	1

1 - low, 2 - medium, 3 - high, '-"- no correlation

24EM401 **EMPLOYABILITY SKILLS II**
Offered by English & Maths (Common to ALL branches) (IV Sem.)

L	T	P	C
0	0	2	1

COURSE OBJECTIVES:
- To understand the problems on surds and indices
- To make the students to clear the concepts of ratio and proportion
- To assess the capability of data assimilation
- To impart knowledge on the importance of resume preparation
- To inculcate the skills required to face interview
- To expose the students for mock interview

UNIT I SURDS AND INDICES 6

Laws of indices- understanding Surds and indices, simplifying surds, rationalizing denominators, different types of surds- Conversion of surds to Indices.

UNIT II RATIO AND PROPORTION 6

Basic Concepts-Definition –Properties of Ratio-Types of Ratio and Proportion- Problems-Applications.

UNIT III DATA INTERPRETATION 6

Introduction- Bar Chart-Pie chart –Tabular form- Histogram –Line graph-Data Sufficiency

UNIT IV RESUME PREPARATION 6

Important components of resume – Types of resumes – Individual resume – cover letter preparation.

UNIT V INTERVIEW SKILLS 6

Interview – preparation – common mistakes in interviews – Do's and Don'ts – Mock Interview.

TOTAL PERIODS: 30

COURSE OUTCOMES:
After the completion of the course, students will be able to
CO1: Acquire the skill on quantitative ability and logical reasoning
CO2: Improve the data assimilation skill
CO3: Develop career competency
CO4: Improve the skill on document preparation
CO5: Enhance the skill on facing interviews
CO6: Know the Do's and Don'ts in the interview process

TEXT BOOK: NIL

REFERENCES:

[1] R. S. Aggarwal, *Quantitative Aptitude*, Revised and Enlarged edition, S. Chand Publication, 2017.
[2] P. A. Anand, *Wiley's Quantitative Aptitude*, 1st Edition, 2015.
[3] Mohan Rao, *Quantitative Aptitude*, SciTech Publications India Pvt. Ltd., Chennai, 2017.
[4] Meenakshi Raman and Sangeetha Sharma, *Technical Communication: English Skills for Engineers*, Oxford University Press, 4th Edition, 2022.
[5] Dr. P. C. Das, *Applied English Grammar and Composition,* New Central Book Agency, Howrah, 2019.

CO's-PO's & PSO's MAPPING

COs	POs												PSOs		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
1	1	1	1	1	-	-	-	-	1	-	1	1	-	-	-
2	1	1	1	1	-	-	-	-	1	-	1	1	-	-	-
3	1	1	1	1	-	-	-	-	1	-	1	1	-	-	-
4	-	-	-	-	-	-	-	2	3	3	-	1	-	-	-
5	-	-	-	-	-	-	-	2	3	3	-	1	-	-	-
6	-	-	-	-	-	-	-	2	3	3	-	1	-	-	-
Avg.	0.5	0.5	0.5	0.5	-	-	-	1	2	1.5	0.5	1	-	-	-

1 - low, 2 - medium, 3 - high, '-"- no correlation